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## 0.1 Hilbert Space

- $\langle \phi | \psi \rangle \langle \phi | \psi \rangle = \langle \psi | \phi \rangle \langle \psi | \phi \rangle^*$ .
- $\langle \psi | \psi \rangle \geq 0$ ; equality iff  $|\psi\rangle = 0$ .
- $\langle \phi | \alpha \psi_1 + \beta \psi_2 \rangle \langle \phi | \alpha \psi_1 + \beta \psi_2 \rangle = \alpha \langle \phi | \psi_1 \rangle \langle \phi | \psi_1 \rangle + \beta \langle \phi | \psi_2 \rangle \langle \phi | \psi_2 \rangle$ .

**Completeness:**  $\sum_n |n\rangle \langle n| = \mathbb{I}$

**Schwarz inequality:**  $|\langle \phi | \psi \rangle|^2 \leq \langle \phi | \phi \rangle \langle \psi | \psi \rangle$

## 0.2 Operators

**Hermitian (self-adjoint):**  $\hat{A} = \hat{A}^\dagger$ ; real eigenvalues, orthogonal eigenstates. Observables are Hermitian.

**Unitary:**  $\hat{U}^\dagger \hat{U} = \hat{U} \hat{U}^\dagger = \mathbb{I}$ ; preserves inner products. Time-evolution operator  $U(t) = e^{-i\hat{H}t/\hbar}$  is unitary.

**Wigner's theorem:** Symmetry preserving transition probabilities  $|\langle \phi | \psi \rangle|$  is unitary or anti-unitary.

**Projection:**  $\hat{P}_n = |n\rangle \langle n|$ ,  $\hat{P}_n^2 = \hat{P}_n$ ,  $\hat{P}_n^\dagger = \hat{P}_n$ .

**Trace:**  $\text{tr}(\hat{A}) = \sum_n \langle n | \hat{A} | n \rangle$ ; cyclic:  $\text{tr}(\hat{A}\hat{B}\hat{C}) = \text{tr}(\hat{C}\hat{A}\hat{B})$ .

**Operator equality from expectations:**  $\langle a | A | a \rangle = \langle a | B | a \rangle \quad \forall |a\rangle \Rightarrow A = B$ . *Proof:* Use  $|\psi\rangle = |a\rangle + |b\rangle$  and  $|\psi\rangle = |a\rangle + i|b\rangle$  to isolate off-diagonal matrix elements.

## 0.3 Operator Identities

**Translation:**  $e^{-i\hat{p}a/\hbar} f(\hat{x}) e^{i\hat{p}a/\hbar} = f(\hat{x} - a)$ . **Scaling:** If  $U_\lambda \psi(x) = \lambda^{1/2} \psi(\lambda x)$ , then  $U_\lambda \hat{x} U_\lambda^\dagger = \hat{x}/\lambda$ ,  $U_\lambda \hat{p} U_\lambda^\dagger = \lambda \hat{p}$ .

## 0.4 Commutators

Properties:

$$\begin{aligned}
 [\hat{A}, \hat{B}] &= -[\hat{B}, \hat{A}], & [\hat{A}, \hat{B}\hat{C}] &= [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}], \\
 [\hat{A}\hat{B}, \hat{C}] &= \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B}, \\
 [\hat{A}, [\hat{B}, \hat{C}]] &+ [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0 \quad (\text{Jacobi}), \\
 [\hat{A}, \hat{A}] &= 0, & [\hat{A}, e^{\hat{C}}] &= 0, & [f(\hat{A}), \hat{A}] &= 0, \\
 \text{If } [[\hat{A}, \hat{B}], \hat{B}] &= 0 \text{ then } [\hat{A}, f(\hat{B})] &= [\hat{A}, \hat{B}] f'(\hat{B}), \\
 [\hat{p}, F(\hat{x})] &= -i\hbar F'(\hat{x}), & [\hat{x}, G(\hat{p})] &= i\hbar G'(\hat{p}), \\
 \text{Example: } [\hat{L}_i, \hat{L}_j] &= i\hbar \epsilon_{ijk} \hat{L}_k, & [\hat{H}, \hat{L}] &= 0.
 \end{aligned}$$

## 0.5 Canonical Quantization

$$[\hat{x}, \hat{p}] = i\hbar, \quad [\hat{x}_i, \hat{p}_j] = i\hbar \delta_{ij}$$

Generalised uncertainty:  $\sigma_A^2 \sigma_B^2 \geq \frac{1}{4} |\langle [\hat{A}, \hat{B}] \rangle|^2$

## 0.6 Useful Identities & Integrals

**Levi-Civita contraction:**  $\epsilon_{ijk} \epsilon_{imn} = \delta_{jm} \delta_{kn} - \delta_{jn} \delta_{km}$ .

**Baker–Campbell–Hausdorff (Hadamard):**

$$e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}] + \frac{1}{2!} [\hat{A}, [\hat{A}, \hat{B}]] + \dots$$

If  $[\hat{A}, [\hat{A}, \hat{B}]] = 0$ :  $e^{\hat{A}} \hat{B} e^{-\hat{A}} = \hat{B} + [\hat{A}, \hat{B}]$ .

**Commutator with functions:**  $[\hat{p}_i, f(r)] = -i\hbar \frac{x_i}{r} f'(r)$ .

**Operator splitting:**  $\hat{A}\hat{B} = \frac{1}{2}[\hat{A}, \hat{B}] + \frac{1}{2}\{\hat{A}, \hat{B}\}$ .

# 1 Quantum Dynamics

## 1.1 Schrödinger Equation

**Time-dependent:**

$$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle$$

**Time-independent:**  $\hat{H} |E_n\rangle = E_n |E_n\rangle$ . General solution:

$$|\Psi(t)\rangle = \sum_n c_n e^{-iE_n t/\hbar} |E_n\rangle$$

## 1.2 Pictures of Quantum Mechanics

**Schrödinger picture:** states evolve, operators fixed.

$$|\Psi_S(t)\rangle = \hat{U}(t) |\Psi_S(0)\rangle, \quad \hat{U}(t) = e^{-i\hat{H}t/\hbar}$$

**Heisenberg picture:** operators evolve, states fixed.

$$\hat{A}_H(t) = \hat{U}^\dagger(t) \hat{A}_S \hat{U}(t), \quad i\hbar \frac{d\hat{A}_H}{dt} = [\hat{A}_H, \hat{H}] + i\hbar \frac{\partial \hat{A}_S}{\partial t}$$

**Key examples** (free particle & HO,  $\partial_t \hat{A}_S = 0$ ):

$$\begin{aligned}
 \dot{\hat{x}}_H &= \frac{\hat{p}_H}{m} & \dot{\hat{p}}_H &= -\left. \frac{\partial V}{\partial x} \right|_{x=\hat{x}_H} \\
 \dot{\hat{a}}_H &= -i\omega \hat{a}_H & \Rightarrow \hat{a}_H(t) &= \hat{a}_H(0) e^{-i\omega t}
 \end{aligned}$$

The last result means the HO in the Heisenberg picture is a classical oscillator for the ladder operators. A coherent state  $|\alpha\rangle$  with  $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$  evolves as  $|\alpha(t)\rangle = |e^{-i\omega t}\alpha\rangle$  — a fact that follows directly from  $\hat{a}_H(t) = \hat{a} e^{-i\omega t}$ .

**Interaction picture:**  $\hat{H} = \hat{H}_0 + V(t)$ .

$$|\Psi_I(t)\rangle = e^{i\hat{H}_0 t/\hbar} |\Psi_S(t)\rangle, \quad \hat{A}_I(t) = e^{i\hat{H}_0 t/\hbar} \hat{A}_S e^{-i\hat{H}_0 t/\hbar}$$

$$i\hbar \frac{\partial}{\partial t} |\Psi_I(t)\rangle = V_I(t) |\Psi_I(t)\rangle$$

## 1.3 Ehrenfest Theorem

$$\frac{d}{dt} \langle \hat{A} \rangle = \frac{1}{i\hbar} \langle [\hat{A}, \hat{H}] \rangle + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle$$

$$m \frac{d}{dt} \langle \hat{x} \rangle = \langle \hat{p} \rangle, \quad \frac{d}{dt} \langle \hat{p} \rangle = -\left\langle \frac{\partial V}{\partial x} \right\rangle$$

## 1.4 Virial Theorem

$$2 \langle T \rangle = \langle \mathbf{r} \cdot \nabla V \rangle$$

- $V \propto r^n$ :  $2 \langle T \rangle = n \langle V \rangle$ .
- Harmonic Osc ( $n = 2$ ):  $\langle T \rangle = \langle V \rangle = E/2$ .
- Coulomb ( $n = -1$ ):  $2 \langle T \rangle = -\langle V \rangle$ , so  $E = -\langle T \rangle = \langle V \rangle/2$ .

## 1.5 Probability Current & Continuity

$$\rho = |\Psi|^2, \quad \mathbf{j} = \frac{-i\hbar}{2m} (\Psi^* \nabla \Psi - \Psi \nabla \Psi^*) = \frac{\hbar}{m} \Im(\Psi^* \nabla \Psi)$$

With EM field  $\mathbf{A}$ :  $\mathbf{j} = (\hbar/m) \Im(\Psi^* \nabla \Psi) - \frac{q}{mc} \mathbf{A} |\Psi|^2$ .

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0$$

## 2 One-Dimensional Systems

### 2.1 Position & Momentum Bases

$$\langle x|x'\rangle = \delta(x-x'), \quad \langle p|p'\rangle = 2\pi\hbar\delta(p-p')$$

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}$$

$$\psi(\mathbf{x}, t) = \frac{1}{L^{3/2}} \exp\left(\frac{i\mathbf{p} \cdot \mathbf{x}}{\hbar} - \frac{iEt}{\hbar}\right)$$

### 2.2 Free Particle

$$\nabla^2\psi(x) = -\frac{2m}{\hbar^2}\psi(x), \quad \mathbf{k}^2 = \frac{2mE}{\hbar^2} = \frac{\mathbf{p}^2}{2m}, \quad k_i = \frac{2\pi}{L}n_i$$

$$\Psi(x, t) = \frac{1}{\sqrt{2\pi\hbar}} \int \tilde{\psi}(p) e^{i(px - p^2 t/2m)/\hbar} dp$$

$$\psi(x) = \frac{1}{L^{1/2}} e^{ik \cdot x}, \quad E = \frac{\mathbf{p}^2}{2m} = \frac{\hbar^2 \mathbf{k}^2}{2m}, \quad \tilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int e^{-ipx/\hbar} \psi(x) dx$$

### 2.3 Infinite Square Well

$V = 0$  for  $0 < x < L$ ,  $V = \infty$  otherwise.

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}, \quad \psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad n = 1, 2, \dots$$

### 2.4 Delta-Function Potential

$V(x) = -\alpha \delta(x)$ ,  $\alpha > 0$ .

**Bound state** ( $E < 0$ , unique):  $E = -\frac{m\alpha^2}{2\hbar^2}$ ,  $\psi(x) = \frac{\sqrt{m\alpha}}{\hbar} e^{-m\alpha|x|/\hbar^2}$ .

**Scattering:** Matching conditions:  $\psi$  continuous,  $\psi'$  discontinuity  $\Delta\psi' = -\frac{2m\alpha}{\hbar^2}\psi(0)$ .

## 3 Harmonic Oscillator

### 3.1 Algebraic (Ladder Operator) Solution

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} u_E(x) + \frac{1}{2} m \omega^2 x^2 u_E(x) = E u_E(x)$$

**Ladder operator definitions**

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \left( \hat{x} + \frac{i\hat{p}}{m\omega} \right), \quad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left( \hat{x} - \frac{i\hat{p}}{m\omega} \right)$$

Inverse:

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger), \quad \hat{p} = i\sqrt{\frac{m\omega\hbar}{2}} (\hat{a}^\dagger - \hat{a})$$

$$[\hat{a}, \hat{a}^\dagger] = 1, \quad \hat{H} = \hbar\omega \left( \hat{a}^\dagger \hat{a} + \frac{1}{2} \right)$$

**Action on Fock states:**

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle, \quad \hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$$

$$E_n = \hbar\omega \left( n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots$$

### 3.2 Ground State

$$\hat{a}|0\rangle = 0 \text{ gives } \psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega x^2/2\hbar}.$$

$$\text{General: } |n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}} |0\rangle.$$

### 3.3 Ladder Operator Calculation Tricks

Express powers of  $\hat{x}$  via  $\hat{a}, \hat{a}^\dagger$  with  $\ell_0 = \sqrt{\hbar/(2m\omega)}$ :

$$\hat{x} = \ell_0(\hat{a} + \hat{a}^\dagger), \quad \hat{x}^2 = \ell_0^2(\hat{a}^2 + \hat{a}^{\dagger 2})$$

$$\hat{x}^2 = \ell_0^2(\hat{a}^2 + (\hat{a}^\dagger)^2 + 2\hat{n} + 1)$$

$$\text{Hence } \langle n|\hat{x}^2|n\rangle = \ell_0^2(2n+1) = \frac{\hbar}{2m\omega}(2n+1).$$

$\hat{x}^4 = \ell_0^4(\hat{a} + \hat{a}^\dagger)^4$ : only keep terms with equal  $\hat{a}, \hat{a}^\dagger$  count for diagonal:

$$\langle n|\hat{x}^4|n\rangle = \ell_0^4(6n^2 + 6n + 3)$$

**Normal-ordering trick:** Expand, commute all  $\hat{a}$  to the right using  $[\hat{a}, \hat{a}^\dagger] = 1$ .

## 4 Symmetry

### 4.1 Parity

$|\Psi_{\text{total}}\rangle = |\psi_{\text{space}}\rangle \otimes |\chi_{\text{spin}}\rangle$ ,  $\hat{\Pi}|\mathbf{r}\rangle = |-\mathbf{r}\rangle$ ,  $\hat{\Pi}^2 = \mathbb{I}$ ; eigenvalues  $\pm 1$ .  
If  $[\hat{H}, \hat{\Pi}] = 0$ , eigenstates can be chosen with definite parity.

$$\hat{\Pi}\hat{x}\hat{\Pi} = -\hat{x}, \quad \hat{\Pi}\hat{p}\hat{\Pi} = -\hat{p}, \quad \hat{\Pi}\hat{\mathbf{L}}\hat{\Pi} = +\hat{\mathbf{L}}.$$

Hydrogen:

$$\Pi\psi(\mathbf{r}) = \psi(-\mathbf{r}), \quad (r, \theta, \phi) \mapsto (r, \pi - \theta, \phi + \pi),$$

$$\Pi[R_{n\ell}Y_{\ell m}] = R_{n\ell}\Pi Y_{\ell m}, \quad e^{im(\phi+\pi)} = (-1)^m e^{im\phi},$$

$$P_\ell^m(\cos(\pi - \theta)) = (-1)^{\ell-m} P_\ell^m(\cos \theta)$$

$$\Pi Y_{\ell m} = (-1)^{\ell-m} (-1)^m Y_{\ell m} = (-1)^\ell Y_{\ell m} \Rightarrow \Pi \psi_{n\ell m} = (-1)^\ell \psi_{n\ell m}$$

E1 selection rule:  $\Delta\ell = \pm 1$ .

States with odd  $\ell$  (p, f, h, ...) have odd parity ( $-1$ ).

States with even  $\ell$  (s, d, g, ...) have even parity ( $+1$ ).

### 4.2 Time Reversal

$\hat{\Theta}$  is *anti-unitary* (antilinear + unitary):  $\hat{\Theta}(c|\psi\rangle) = c^*\hat{\Theta}|\psi\rangle$ , preserves inner products up to complex conjugation.

$$\hat{\Theta}^2 = \begin{cases} +1 & \text{for integer spin (bosons),} \\ -1 & \text{for half-integer spin (fermions).} \end{cases}$$

For spin-0 particles:  $(\hat{\Theta}\Psi)(x) = \Psi^*(x)$ .

For spin- $\frac{1}{2}$  particles:  $(\hat{\Theta}\Psi)(x) = \begin{pmatrix} -\psi_\downarrow^*(x) \\ \psi_\uparrow^*(x) \end{pmatrix}$ , or  $\hat{\Theta} = -i\sigma_y \mathcal{K}$ , where  $\mathcal{K}$  is complex conjugation.

Transformations under time reversal:

$$\hat{\Theta}\hat{x}\hat{\Theta}^{-1} = \hat{x}, \quad \hat{\Theta}\hat{p}\hat{\Theta}^{-1} = -\hat{p}, \quad \hat{\Theta}\hat{\mathbf{J}}\hat{\Theta}^{-1} = -\hat{\mathbf{J}}, \quad \hat{\Theta}\hat{\mathbf{S}}\hat{\Theta}^{-1} = -\hat{\mathbf{S}}$$

$$\hat{\Theta}\hat{\mathbf{B}}\hat{\Theta}^{-1} = -\hat{\mathbf{B}}, \quad \hat{\Theta}i\hat{\Theta}^{-1} = -i$$

For half-integer spin with  $[\hat{H}, \hat{\Theta}] = 0$ , every energy level is at least doubly degenerate. **Composite Systems:** Parity is a multiplicative quantum number. For an  $N$ -particle system with relative orbital angular momenta  $l_i$  and intrinsic particle parities  $\eta_i$ :

$$\pi_{\text{total}} = \prod_{i=1}^N \eta_i (-1)^{l_i}$$

*Note:* Fermion-antifermion pairs inherently carry a relative intrinsic parity of  $\eta_f \eta_{\bar{f}} = -1$ .

**Matrix Elements & Selection Rules:** For an operator  $\hat{O}$  with definite parity  $\pi_O$  (where  $\hat{\Pi}^\dagger \hat{O} \hat{\Pi} = \pi_O \hat{O}$ ), the expectation value or transition matrix element dictates Laporte's rule:

$$\langle \psi_f | \hat{O} | \psi_i \rangle \neq 0 \Rightarrow \pi_f \cdot \pi_O \cdot \pi_i = +1$$

*Example:* The electric dipole operator  $\hat{\mathbf{d}} = e\hat{\mathbf{r}}$  has odd parity ( $\pi_O = -1$ ), necessitating  $\pi_f = -\pi_i$  (e.g.,  $\Delta l = \pm 1$ ).

### 4.3 Translation

$$\hat{T}(\mathbf{a}) = e^{-i\hat{\mathbf{p}} \cdot \mathbf{a}/\hbar}, \quad \hat{T}(\mathbf{a})|\mathbf{r}\rangle = |\mathbf{r} + \mathbf{a}\rangle$$

$\hat{p}$  generates translations:  $[\hat{p}, \hat{T}(\mathbf{a})] = 0$ .

$$\hat{R}(\hat{n}, \theta) = e^{-i\hat{\mathbf{J}} \cdot \hat{n} \theta / \hbar}$$

$SO(3)$ : group of  $3 \times 3$  real orthogonal matrices with  $\det = +1$ .

$SU(2)$ : group of  $2 \times 2$  unitary matrices with  $\det = 1$ ; double covers  $SO(3)$ .

Wigner  $D$ -matrices:

$$D_{m'm}^{(j)}(\alpha, \beta, \gamma) = \langle j, m' | e^{-i\alpha \hat{J}_z / \hbar} e^{-i\beta \hat{J}_y / \hbar} e^{-i\gamma \hat{J}_z / \hbar} | j, m \rangle.$$

Explicit  $\ell = 1$  matrices (basis  $m = +1, 0, -1$ ):

$$L_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad L_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -i & 0 \\ i & 0 & -i \\ 0 & i & 0 \end{pmatrix}$$

Cayley-Hamilton for  $j = 1$ :  $(J_y/\hbar)^3 = J_y/\hbar$ , so:

$$e^{-i\beta J_y/\hbar} = \mathbb{I} - (J_y/\hbar)^2(1 - \cos \beta) - i(J_y/\hbar) \sin \beta$$

Ladder operator trick:  $L_x^2 - L_y^2 = \frac{1}{2}(L_+^2 + L_-^2)$  (shifts  $m$  by  $\pm 2$ ).

## 5 Angular Momentum

### 5.1 General Angular Momentum Algebra

Commutation relations (defining relation of  $\mathfrak{su}(2)$ ):

$$[\hat{J}_i, \hat{J}_j] = i\hbar \epsilon_{ijk} \hat{J}_k$$

$\hat{J}^2 = \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2$  commutes with all  $\hat{J}_i$ :  $[\hat{J}^2, \hat{J}_i] = 0$ .

Ladder operators for angular momentum

$$\hat{J}_{\pm} = \hat{J}_x \pm i\hat{J}_y \quad \Leftrightarrow \quad \hat{J}_x = \frac{\hat{J}_+ + \hat{J}_-}{2}, \quad \hat{J}_y = \frac{\hat{J}_+ - \hat{J}_-}{2i}$$

Key commutators:

$$[\hat{J}_z, \hat{J}_{\pm}] = \pm \hbar \hat{J}_{\pm}, \quad [\hat{J}_+, \hat{J}_-] = 2\hbar \hat{J}_z$$

$$\hat{J}^2 = \hat{J}_- \hat{J}_+ + \hat{J}_z^2 + \hbar \hat{J}_z = \hat{J}_+ \hat{J}_- + \hat{J}_z^2 - \hbar \hat{J}_z$$

Key identity (spin-orbit):

$$\vec{L} \cdot \vec{S} = \frac{1}{2}(\hat{J}^2 - \hat{L}^2 - \hat{S}^2)$$

with  $\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$ . In the coupled basis  $|n, \ell, s, j, m_j\rangle$  this is diagonal:

$$\langle \vec{L} \cdot \vec{S} \rangle = \frac{\hbar^2}{2}[j(j+1) - \ell(\ell+1) - s(s+1)].$$

Ladder decomposition:

$$\vec{L} \cdot \vec{S} = \hat{L}_z \hat{S}_z + \frac{1}{2}(\hat{L}_+ \hat{S}_- + \hat{L}_- \hat{S}_+)$$

Spectrum:

$$\hat{J}^2 |j, m\rangle = \hbar^2 j(j+1) |j, m\rangle, \quad \hat{J}_z |j, m\rangle = \hbar m |j, m\rangle$$

$$\hat{J}_{\pm} |j, m\rangle = \hbar \sqrt{j(j+1) - m(m \pm 1)} |j, m \pm 1\rangle$$

$j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ ;  $m = -j, -j+1, \dots, j$ . Derived from  $\hat{J}_x^2 + \hat{J}_y^2 \geq 0$  (bounds  $m$ ) and ladder truncation ( $\hat{J}_{\pm} |j, j\rangle = 0$ ).

### 5.2 Orbital Angular Momentum & Spherical Harmonics

$\hat{\mathbf{L}} = \mathbf{r} \times \hat{\mathbf{p}}$ ; integer  $\ell$  only ( $\ell = 0, 1, 2, \dots$ ).

$$\hat{L}^2 Y_{\ell}^m = \hbar^2 \ell(\ell+1) Y_{\ell}^m, \quad \hat{L}_z Y_{\ell}^m = \hbar m Y_{\ell}^m$$

## 6 Central Potentials & Hydrogen Atom

Separation:  $\psi_{n\ell m} = R_{n\ell}(r) Y_{\ell}^m(\theta, \phi)$

$$Y_{\ell}^m = (-1)^m \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-m)!}{(\ell+m)!}} P_{\ell}^m(\cos \theta) e^{im\phi}$$

$$R_{n\ell}(r) = \sqrt{\left(\frac{2}{na_0}\right)^3 \frac{(n-\ell-1)!}{2n(n+\ell)!}} e^{-r/na_0} \left(\frac{2r}{na_0}\right)^{\ell} L_{n-\ell-1}^{2\ell+1}\left(\frac{2r}{na_0}\right)$$

Radial equation ( $u \equiv rR$ ,  $u(0) = 0$ ,  $\int_0^{\infty} |u|^2 dr = 1$ ):

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[ V(r) + \frac{\hbar^2 \ell(\ell+1)}{2mr^2} \right] u = Eu$$

### 6.1 Spherical Bessel Functions ( $V = 0$ , $k^2 = 2mE/\hbar^2$ )

$R_{\ell}(r) = A j_{\ell}(kr) + B n_{\ell}(kr)$ ;  $j_{\ell} \sim r^{\ell}$  (regular),  $n_{\ell} \sim r^{-(\ell+1)}$  (singular).

Infinite spherical well ( $V = 0$ ,  $r < a$ ;  $V = \infty$ ,  $r \geq a$ ):

B.C.:  $j_{\ell}(k_{n\ell}a) = 0$ ; let  $z_{n\ell}$  be the  $n$ -th positive zero of  $j_{\ell}$ :

$$E_{n\ell} = \frac{\hbar^2 z_{n\ell}^2}{2ma^2}, \quad z_{10} \approx \pi, \quad z_{11} \approx 4.49, \quad z_{20} \approx 2\pi$$

### 6.2 Hydrogen Atom ( $V = -e^2/4\pi\epsilon_0 r$ , $a_0 = 4\pi\epsilon_0 \hbar^2/me^2$ )

$$E_n = -\frac{13.6 \text{ eV}}{n^2} = -\frac{e^2}{8\pi\epsilon_0 a_0 n^2}, \quad n = 1, 2, \dots; \quad \ell = 0, \dots, n-1;$$

$m = -\ell, \dots, \ell$

Degeneracy:  $n^2$  (no spin),  $2n^2$  (with spin). Virial:  $E_n = -\langle T \rangle_n = \frac{1}{2} \langle V \rangle_n$ .

At origin:  $|\psi_{n\ell m}(0)|^2 = 0$  ( $\ell \geq 1$ );  $|\psi_{n00}(0)|^2 = \frac{1}{\pi n^3 a_0^3}$ .

E1 selection rules:  $\Delta \ell = \pm 1$ ;  $\Delta m = 0, \pm 1$ ;  $\Delta n$  unrestricted.

Dipole element:

$$\langle 2, 1, 0 | z | 1, 0, 0 \rangle = \frac{128\sqrt{2}}{243} a_0. \quad \langle 2, 1, \pm 1 | (x \pm iy) | 1, 0, 0 \rangle = \mp \frac{128\sqrt{2}}{243} a_0.$$

### 6.3 Hydrogen Basis & Angular Momentum

Hamiltonian Components:

$$H_0 = \frac{p^2}{2m} - \frac{e^2}{4\pi\epsilon_0 r} \rightarrow \left( -\frac{\hbar^2}{2\mu} \nabla^2 - \frac{e^2}{4\pi\epsilon_0 r} \right) \psi(r, \theta, \phi) = E\psi(r, \theta, \phi)$$

$$H_{FS} \approx \xi(r) \mathbf{L} \cdot \mathbf{S} = \frac{\xi(r)}{2} (J^2 - L^2 - S^2)$$

$$H_Z = \frac{\mu_B}{\hbar} (L_z + 2S_z) B_{ext} \quad (\mathbf{B} = B_{ext} \hat{z})$$

1. Uncoupled Basis (Product Space):  $|n, \ell, m_{\ell}, m_s\rangle$

- CSCO:  $\{H_0, L^2, S^2, L_z, S_z\}$
- Good Q.N.:  $n, \ell, s = \frac{1}{2}, m_{\ell}, m_s$

2. Coupled Basis (Total Ang. Mom.):  $|n, \ell, j, m_j\rangle$

- CSCO:  $\{H_0, L^2, S^2, J^2, J_z\}$  where  $\mathbf{J} = \mathbf{L} + \mathbf{S}$
- Good Q.N.:  $n, \ell, s = \frac{1}{2}, j, m_j$
- Allowed  $j$ :  $|\ell - s| \leq j \leq \ell + s$
- Relation:  $[\mathbf{L} \cdot \mathbf{S}, \mathbf{J}] = 0$ .

Basis Transformation (Clebsch-Gordan):

$$|j, m_j\rangle = \sum_{m_{\ell}, m_s} C_{\ell m_{\ell} s m_s}^{j m_j} |\ell, m_{\ell}; s, m_s\rangle$$

Selection Rules:  $m_j = m_{\ell} + m_s$ , triangle rule for  $(\ell, s, j)$ .

Perturbation Strategy:

- If  $B_{ext} \rightarrow 0$ : Use **Coupled**.  $E_{FS}^{(1)} = \langle j m_j | H_{FS} | j m_j \rangle$ .
- If  $B_{ext} \rightarrow \infty$ : Use **Uncoupled**.  $E_Z^{(1)} = \mu_B B(m_{\ell} + 2m_s)$ .
- If  $B_{ext} \sim H_{FS}$ : Diagonalize matrix of  $H_{FS} + H_Z$  in Coupled basis (usually).

## 6.4 Hydrogen-like Atoms ( $Z > 1$ )

Replace  $e^2 \rightarrow Ze^2$ :  $a_0 \rightarrow a_0/Z$ ,  $E_n \rightarrow -Z^2 \cdot 13.6 \text{ eV}/n^2$ .

$\langle r \rangle_{n=1} \sim a_0/Z$ .

**Reduced mass:**  $m_e \rightarrow \mu = \frac{m_e M_{\text{nuc}}}{m_e + M_{\text{nuc}}}$ ;  $a_0 \rightarrow a_\mu = a_0 \frac{m_e}{\mu}$ ,  $Ry \rightarrow Ry \frac{\mu}{m_e}$ .

**Fine-structure constant:**  $\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137}$ . Non-relativistic regime valid when  $Z\alpha \ll 1$ .

## 6.5 Radial Equation

$V(r) = -e^2/(4\pi\epsilon_0 r)$ . With  $u = rR$ ,  $\rho = r/(na_0)$ :

$$\frac{d^2 u}{d\rho^2} = \left[ 1 - \frac{2n}{\rho} + \frac{\ell(\ell+1)}{\rho^2} \right] u$$

Solution:  $u_{n\ell}(\rho) = \rho^{\ell+1} e^{-\rho} L_{n-\ell-1}^{2\ell+1}(2\rho)$ .

$$\psi_{100} = \frac{1}{\sqrt{\pi}} \left( \frac{1}{a_0} \right)^{3/2} e^{-r/a_0}, \quad \psi_{200} = \frac{1}{4\sqrt{2\pi}} \left( \frac{1}{a_0} \right)^{3/2} \left( 2 - \frac{r}{a_0} \right) e^{-r/2a_0}$$

$$\psi_{210} = \frac{1}{4\sqrt{2\pi}} \left( \frac{1}{a_0} \right)^{3/2} \frac{r}{a_0} e^{-r/2a_0} \cos \theta, \quad \psi_{211} = -\frac{1}{8\sqrt{\pi}} \left( \frac{1}{a_0} \right)^{3/2} \frac{r}{a_0} e^{-r/2a_0} \sin \theta e^{i\phi}$$

## 6.6 Expectation Values

$\langle r^k \rangle_{n\ell}$  for hydrogen:

$$\begin{aligned} \langle r \rangle &= \frac{a_0}{2} [3n^2 - \ell(\ell+1)], \quad \langle r^2 \rangle = \frac{a_0^2 n^2}{2} [5n^2 + 1 - 3\ell(\ell+1)] \\ \langle 1/r \rangle &= \frac{1}{n^2 a_0}, \quad \langle 1/r^2 \rangle = \frac{1}{n^3 a_0^2 (\ell + \frac{1}{2})}, \quad \langle 1/r^3 \rangle = \frac{1}{n^3 a_0^3 \ell (\ell + \frac{1}{2}) (\ell + 1)} \quad (\ell \geq 1) \end{aligned}$$

## 7 Addition of Angular Momentum

### 7.1 Tensor Products & CSCO

$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$ ;  $\dim = (2j_1 + 1)(2j_2 + 1)$ .

**Uncoupled basis**  $|j_1, m_1; j_2, m_2\rangle$ : eigenstates of  $\{\hat{J}_1^2, \hat{J}_{1z}, \hat{J}_2^2, \hat{J}_{2z}\}$ .

**Total:**  $\hat{\mathbf{J}} = \hat{\mathbf{J}}_1 \otimes \mathbb{1}_2 + \mathbb{1}_1 \otimes \hat{\mathbf{J}}_2$ ;  $J$  ranges  $|j_1 - j_2| \leq J \leq j_1 + j_2$ .

**Decomposition:**

$$j_1 \otimes j_2 = \bigoplus_{J=|j_1-j_2|}^{j_1+j_2} J \quad (\text{integer steps})$$

**Dimension check:**  $(2j_1 + 1)(2j_2 + 1) = \sum_{J=|j_1-j_2|}^{j_1+j_2} (2J + 1)$ .

**Coupled basis**  $|J, M; j_1, j_2\rangle$ : eigenstates of  $\{\hat{J}^2, \hat{J}_z, \hat{J}_1^2, \hat{J}_2^2\}$ , with  $M = m_1 + m_2$ .

**Highest weight:**

$$|J_{\max}, M_{\max}\rangle = |j_1, j_1\rangle |j_2, j_2\rangle, \quad J_{\max} = j_1 + j_2.$$

**Lowering:**

$$J_- = J_{1-} + J_{2-}$$

generates the  $J_{\max}$  multiplet.

**Orthogonal remainder:**

The highest state of  $J_{\max} - 1$  is the  $M = J_{\max} - 1$  state orthogonal to the  $J_{\max}$  ladder.

**Examples:**  $\frac{1}{2} \otimes 1 = \frac{1}{2} \oplus \frac{3}{2}$ ,  $1 \otimes 1 = 0 \oplus 1 \oplus 2$ .

Moreover,  $\hat{J}^2$  is *not* diagonal in the uncoupled basis because  $\hat{J}^2 = (\hat{\mathbf{J}}_1 + \hat{\mathbf{J}}_2)^2$

contains the term  $2\hat{\mathbf{J}}_1 \cdot \hat{\mathbf{J}}_2$ . Rotation in product space:  $U(R) = U_1(R) \otimes U_2(R)$ .

## 7.2 Clebsch–Gordan Coefficients

$$|J, M\rangle = \sum_{m_1+m_2=M} \langle j_1, m_1; j_2, m_2 | J, M \rangle |j_1, m_1; j_2, m_2\rangle$$

nonzero if  $|j_1 - j_2| \leq J \leq j_1 + j_2$  and  $M = m_1 + m_2$ .

**Symmetry:**

$$\langle j_1 m_1; j_2 m_2 | J M \rangle |j_1 m_1; j_2 m_2 | J M \rangle = (-1)^{j_1+j_2-J} \langle j_2 m_2; j_1 m_1 | J M \rangle |j_2 m_2; j_1 m_1 | J M \rangle$$

For  $j_1 = j_2$  and  $m_1 = m_2$ , this vanishes if  $J$  is odd.

## 7.3 General Relation

For identical angular momenta:

$$\langle j_0 j_0 m_1 m_2 | J M \rangle = (-1)^{2j_0-J} \langle j_0 j_0 m_2 m_1 | J M \rangle$$

$$\hat{P}_{12} |J, M\rangle = (-1)^{2j_0-J} |J, M\rangle$$

## 7.4 Properties

• **Symmetry Condition:** The state is Symmetric if  $(2j_0 - J)$  is even, and Antisymmetric if odd.

• **Alternating Pattern:**

- $J = 2j_0$  (Max): Symmetric (+1)
- $J = 2j_0 - 1$ : Antisymmetric (−1)

**Exchange Symmetry ( $\hat{P}_{12}$ ) for Identical Particles** ( $j_1 = j_2 \equiv j$ ,  $\mathbf{J} = \mathbf{j}_1 + \mathbf{j}_2$ ): Eigenvalue:  $\hat{P}_{12} |J, M\rangle = (-1)^{2j-J} |J, M\rangle$ . Max  $J$  ( $2j$ ) is always symmetric.

Fermions ( $j \in \mathbb{Z} + \frac{1}{2}$ )  $\implies$  Symmetric: odd  $J$ , Antisymmetric: even  $J$

Bosons ( $j \in \mathbb{Z}$ )  $\implies$  Symmetric: even  $J$ , Antisymmetric: odd  $J$

Spin-1/2 ( $j = \frac{1}{2}$ )  $\implies$  Singlet  $S = 0$  (Anti), Triplet  $S = 1$  (Sym)

## 7.5 Spherical Tensors

A **spherical tensor operator** of rank  $k$  is a set of  $2k + 1$  operators  $T_q^{(k)}$  ( $q = -k, \dots, k$ ) satisfying:

$$[J_z, T_q^{(k)}] = \hbar q T_q^{(k)}, \quad [J_{\pm}, T_q^{(k)}] = \hbar \sqrt{k(k+1) - q(q \pm 1)} T_{q \pm 1}^{(k)}$$

They transform under rotations as angular momentum states of spin  $k$ .

$$U(R) T_q^{(k)} U(R)^\dagger = \sum_{q'} D_{qq'}^{(k)}(R) T_{q'}^{(k)}$$

Under parity:  $\hat{P}^\dagger T_q^{(k)} \hat{P} = (-1)^k T_q^{(k)}$

$$\text{Rank 0: } T_0^{(0)} = -\frac{1}{\sqrt{3}} \mathbf{u} \cdot \mathbf{v} \quad (\text{Ex: } H, \mathbf{r} \cdot \mathbf{p}, \mathbf{S} \cdot \mathbf{L})$$

$$\text{Rank 1: } V_{\pm 1}^{(1)} = \mp \frac{1}{\sqrt{2}} (V_x \pm iV_y), \quad V_0^{(1)} = V_z \quad (\text{Op: } \mathbf{L}, \mathbf{S}, \mathbf{J}, \mathbf{p}, \mathbf{r})$$

$$\mathbf{r}: r_{\pm 1}^{(1)} = \mp \frac{1}{\sqrt{2}} (x \pm iy) = \mp r \sqrt{\frac{4\pi}{3}} Y_1^{\pm 1}, \quad r_0^{(1)} = z = r \sqrt{\frac{4\pi}{3}} Y_1^0$$

$$\mathbf{J}: J_{\pm 1}^{(1)} = \mp \frac{1}{\sqrt{2}} J_{\pm}, \quad J_0^{(1)} = J_z, \quad \mathbf{p}: p_{\pm 1}^{(1)} = \mp \frac{1}{\sqrt{2}} (p_x \pm ip_y), \quad p_0^{(1)} = p_z$$

$$\text{Rank 2: } T_q^{(2)} = \sum_{m_1, m_2} \langle 1, m_1; 1, m_2 | 2, q \rangle u_{m_1}^{(1)} v_{m_2}^{(1)} \quad (\text{Op: } Q_{ij}, \sigma_{ij})$$

$$Q: Q_{\pm 2}^{(2)} = \sqrt{\frac{3}{8}} (x \pm iy)^2, \quad Q_{\pm 1}^{(2)} = \mp \sqrt{\frac{3}{2}} z(x \pm iy), \quad Q_0^{(2)} = \frac{1}{2} (3z^2 - r^2)$$

## Wigner-Eckart Theorem

For a **spherical tensor operator**  $\hat{T}_q^{(k)}$ :

$$\langle j', m' | \hat{T}_q^{(k)} | j, m \rangle = \frac{\langle j' || \hat{T}^{(k)} || j \rangle}{\sqrt{2j'+1}} \langle j, m; k, q | j', m' \rangle$$

All  $m$ -dependence is in the Clebsch–Gordan coefficient; the *reduced matrix element*  $\langle j' || \hat{T}^{(k)} || j \rangle$  is independent of  $m, m', q$ .

For a spherical tensor operator  $T_q^{(k)}$ , the matrix element satisfies:

$$\langle \alpha', j' m' | T_q^{(k)} | \alpha, j m \rangle = 0, \quad \text{unless } m' = q + m$$

- The  $z$ -component of angular momentum is conserved; the operator changes  $m$  by  $q\hbar$ .
- A transition to  $j'$  is forbidden if  $T^{(k)}$  does not supply the required angular momentum.

- For states of definite parity, the transition is allowed only if

$$\pi_{final} = \pi_{op} \times \pi_{initial}$$

- Magnetic dipole (M1) transitions: the operator is a pseudovector ( $\mathbf{L}$ , parity +1); parity must remain unchanged.
- Scalar operators ( $k = 0$ ) cannot change  $j$  or  $m$ ; they connect only states with identical angular momentum quantum numbers.

## 7.6 The Projection Theorem

For matrix elements diagonal in  $j$  ( $j = j'$ ), the Wigner-Eckart theorem simplifies to:

$$\langle \alpha', j, m' | V_q | \alpha, j, m \rangle = \frac{\langle \alpha', j, m | \mathbf{J} \cdot \mathbf{V} | \alpha, j, m \rangle}{\hbar^2 j(j+1)} \langle j, m' | J_q | j, m \rangle$$

where the spherical components of angular momentum are defined as:

$$J_{\pm 1} = \mp \frac{1}{\sqrt{2}} (J_x \pm iJ_y), \quad J_0 = J_z$$

**Decomposition**  $1 \otimes 1 = 0 \oplus 1 \oplus 2$ : scalar  $T_0^{(0)} = -\frac{1}{\sqrt{3}}(\mathbf{W} \cdot \mathbf{V})$ , vector  $\propto \mathbf{W} \times \mathbf{V}$ , rank-2 = traceless symmetric.

$$r_{1i}r_{2j} = \left( \frac{\mathbf{r}_1 \cdot \mathbf{r}_2}{3} \delta_{ij} \right) + \left( \frac{r_{1i}r_{2j} - r_{1j}r_{2i}}{2} \right) + \left( \frac{r_{1i}r_{2j} + r_{1j}r_{2i}}{2} - \frac{\mathbf{r}_1 \cdot \mathbf{r}_2}{3} \delta_{ij} \right)$$

## 8 Selection Rules

### 8.1 Parity Selection Rule

Parity of hydrogen state:  $(-1)^\ell$ . For any parity-odd operator  $\hat{O}$  (e.g.  $\hat{x}, \hat{y}, \hat{z}$ ):

$$\langle n\ell m | \hat{O} | n'\ell' m' \rangle = 0 \quad \text{unless } \ell + \ell' \text{ is odd}$$

**Proof sketch.** Insert  $\hat{\Pi}^\dagger \hat{\Pi} = \mathbb{I}$  on both sides;  $\hat{\Pi} \hat{O} \hat{\Pi}^\dagger = -\hat{O} \Rightarrow$  factor  $(-1)^{\ell+\ell'+1}$ . The integrand  $\psi_{nlm}^* \hat{O} \psi_{n'\ell'm'}$  must be even over all space.

### 8.2 Electric Dipole Selection Rules

For transitions driven by the operator  $\hat{\mathbf{r}}$  (or  $\epsilon \cdot \hat{\mathbf{p}}$  in the dipole approx.):

$$\Delta\ell = \pm 1, \quad \Delta m = 0, \pm 1$$

$\Delta m = 0$  for  $\hat{z}$ -polarisation;  $\Delta m = \pm 1$  for  $\hat{x} \pm i\hat{y}$  (circular).

From Wigner-Eckart: the dipole operator is a rank-1 spherical tensor ( $k = 1$ ), hence  $|\ell - 1| \leq \ell' \leq \ell + 1$  and  $\Delta m = q \in \{0, \pm 1\}$ ; parity forbids  $\Delta\ell = 0$ .

**General Selection Rules** (Non-vanishing of the reduced matrix element):

- m-Conservation:**  $m' = m + q$
- Triangle Inequality:**  $|j - k| \leq j' \leq j + k$  (Vector addition  $\vec{j} + \vec{k} = \vec{j'}$ )
- Parity:**  $\pi_f = \pi_{op} \cdot \pi_i$ . If  $\pi_{op}$  has parity  $(-1)^k$  (e.g., Electric),  $\pi_f = (-1)^k \pi_i$ .

**EM Transitions (Single Electron  $n\ell \rightarrow n'\ell'$ )**

- Parity of EM Operators:** Electric  $E(k) \rightarrow (-1)^k$ , Magnetic  $M(k) \rightarrow (-1)^{k+1}$ .
- Photon Spin:**  $S_\gamma = 1 \Rightarrow \Delta J \neq 0$  for  $0 \rightarrow 0$  transitions.

| Type        | Operator                         | Parity   | Exact Rules ( $\Delta J$ )         | LS Rules ( $\Delta S = 0$ ) |
|-------------|----------------------------------|----------|------------------------------------|-----------------------------|
| E1 (Dipole) | $\vec{r}$ ( $k = 1$ )            | Flip (-) | $0, \pm 1$ (No $0 \rightarrow 0$ ) | $\Delta\ell = \pm 1$        |
| M1 (Dipole) | $\vec{L} + 2\vec{S}$ ( $k = 1$ ) | Same (+) | $0, \pm 1$ (No $0 \rightarrow 0$ ) | $\Delta\ell = 0$            |
| E2 (Quad.)  | $x_i x_j$ ( $k = 2$ )            | Same (+) | $0, \pm 1, \pm 2$                  | $\Delta\ell = 0, \pm 2$     |

## 9 Spin

**Total Ladder Operators:**

$$\hat{S}_\pm = \hat{S}_{1\pm} + \hat{S}_{2\pm} = (\hat{S}_{1x} \pm i\hat{S}_{1y}) + (\hat{S}_{2x} \pm i\hat{S}_{2y})$$

**Action on Tensor Products:** Individual operators act only on their respective subspaces:

$$\hat{S}_{1-} |m_1, m_2\rangle = (\hat{S}_{1-} |m_1\rangle) |m_2\rangle, \quad \hat{S}_{2-} |m_1, m_2\rangle = |m_1\rangle (\hat{S}_{2-} |m_2\rangle)$$

This allows shifting the total  $M$  while preserving the  $J$  multiplet.

## Two Spin- $\frac{1}{2}$ Addition (Singlet & Triplet)

Total spin  $\mathbf{S} = \mathbf{S}_1 + \mathbf{S}_2$ . Eigenvalues:  $S^2 \rightarrow s(s+1)\hbar^2$ ,  $S_z \rightarrow m\hbar$ .

**1. Triplet ( $s = 1$ ):** Symmetric under particle exchange.

$$|1, 1\rangle = |\uparrow\uparrow\rangle \quad (\text{or } |++\rangle)$$

$$|1, 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)$$

$$|1, -1\rangle = |\downarrow\downarrow\rangle \quad (\text{or } |--\rangle)$$

**2. Singlet ( $s = 0$ ):** Antisymmetric under particle exchange.

$$|0, 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

*Note:*  $S = 1$  states are symmetric;  $S = 0$  is antisymmetric. This dictates spatial symmetry for identical fermions (Pauli principle).

### 9.1 Spin- $\frac{1}{2}$ & Pauli Matrices

$\hat{\mathbf{S}} = \frac{\hbar}{2} \boldsymbol{\sigma}$  with

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\sigma_i \sigma_j = \delta_{ij} \mathbb{I} + i\epsilon_{ijk} \sigma_k; \quad \{\sigma_i, \sigma_j\} = 2\delta_{ij} \mathbb{I}.$$

**Eigenvalues for spin- $\frac{1}{2}$ :**

$$\hat{S}^2 |\uparrow\rangle = \frac{3\hbar^2}{4} |\uparrow\rangle, \quad \hat{S}^2 |\downarrow\rangle = \frac{3\hbar^2}{4} |\downarrow\rangle$$

$$\hat{S}_z |\uparrow\rangle = +\frac{\hbar}{2} |\uparrow\rangle, \quad \hat{S}_z |\downarrow\rangle = -\frac{\hbar}{2} |\downarrow\rangle$$

$$e^{-i\theta \hat{n} \cdot \boldsymbol{\sigma}/2} = \cos(\theta/2) \mathbb{I} - i \sin(\theta/2) \hat{n} \cdot \boldsymbol{\sigma}.$$

**Pauli vector identity:**

$$(\vec{\sigma} \cdot \mathbf{a})(\vec{\sigma} \cdot \mathbf{b}) = (\mathbf{a} \cdot \mathbf{b}) \mathbb{I} + i \vec{\sigma} \cdot (\mathbf{a} \times \mathbf{b})$$

Special case:  $(\hat{n} \cdot \vec{\sigma})^2 = \mathbb{I}$ ; eigenvalues of  $\hat{n} \cdot \vec{\sigma}$  are  $\pm 1$ .

Spin- $\frac{1}{2}$  state:  $|\hat{n}, +\rangle = \cos(\theta/2) |\uparrow\rangle + e^{i\phi} \sin(\theta/2) |\downarrow\rangle$ .

$\langle \hat{S} \rangle_{\hat{n},+} = \frac{\hbar}{2} \hat{n}$ ; measurement along  $\hat{n}'$ :  $P(+) = \cos^2(\gamma/2)$  where  $\gamma = \angle(\hat{n}, \hat{n}')$ .

### 9.2 Spin Dynamics

$$\hat{H} = -\boldsymbol{\mu} \cdot \mathbf{B} = -\gamma \hat{\mathbf{S}} \cdot \mathbf{B}.$$

**Larmor precession:** Constant  $\mathbf{B} = B_0 \hat{z}$ :  $\omega_L = \gamma B_0$ ; spin precesses about  $\hat{z}$ .

**Rabi oscillations:**  $\mathbf{B} = B_0 \hat{z} + B_1 [\cos(\omega t) \hat{x} + \sin(\omega t) \hat{y}]$ .

Transition probability (resonance at  $\omega = \omega_0$ ):

$$P_{\uparrow \rightarrow \downarrow}(t) = \frac{\Omega_R^2}{\Omega_R^2 + \delta^2} \sin^2 \left( \frac{\sqrt{\Omega_R^2 + \delta^2}}{2} t \right)$$

where  $\Omega_R = \gamma B_1$  (Rabi frequency),  $\delta = \omega - \omega_0$  (detuning).

## 10 Electromagnetic Interactions

### 10.1 Gauge Invariance

Minimal coupling:  $\hat{p} \rightarrow \boldsymbol{\Pi} = \hat{p} - q\mathbf{A}/c$ .

$$H = \frac{\mathbf{p}^2}{2m} - \frac{q}{2m} (\mathbf{p} \cdot \mathbf{A} + \mathbf{A} \cdot \mathbf{p}) + \frac{q^2}{2m} \mathbf{A}^2 + q\phi$$

Gauge transformation  $\mathbf{A} \rightarrow \mathbf{A} + \nabla\chi$ ,  $\Phi \rightarrow \Phi - \frac{1}{c} \frac{\partial\chi}{\partial t}$  compensated by  $|\psi\rangle \rightarrow e^{iq\chi/\hbar c} |\psi\rangle$ . Physics (probabilities, spectra) unchanged.

### 10.2 Landau Levels

$$\mathbf{B} = B\hat{z}; \quad \mathbf{A}_{\text{Landau}} = (-By, 0, 0), \quad A_{\text{sym}} = \frac{B}{2}(-y, x, 0)$$

$$E_n = \hbar\omega_c \left( n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots, \quad \omega_c = \frac{|q|B}{mc}$$

Each level is macroscopically degenerate ( $\propto B \cdot \text{Area}$ ).

$\mathbf{p}$  is gauge dependent while  $\boldsymbol{\Pi}$  is gauge invariant.



### 10.3 Aharonov-Bohm Effect

Charged particle encircles a flux  $\Phi_B$  (where  $\mathbf{B} = 0$  locally).

- **Scattering (Open Path):** Wavefunction picks up a path-dependent phase:

$$\oint \mathbf{A} \cdot d\mathbf{l} = \int (\nabla \times \mathbf{A}) \cdot d\mathbf{S} = \int \mathbf{B} \cdot d\mathbf{S} = \Phi_B, \quad \Delta\varphi = \frac{q}{\hbar c} \Phi_B$$

$$\psi = \psi_0 e^{i\varphi}, \quad \Delta\varphi = \frac{q}{\hbar c} \oint \mathbf{A} \cdot d\boldsymbol{\ell} = 2\pi \frac{\Phi_B}{(\hbar c/q)}$$

Observable interference depends on  $\Phi_B \pmod{\Phi_0}$ .

- **Bound State (Ring Radius  $R$ ):** Rotational symmetry ( $[\hat{H}, \hat{L}_z] = 0$ ) but broken time-reversal.

$$H = \frac{1}{2mR^2} \left( \hat{L}_z - \frac{q\Phi_B}{2\pi c} \right)^2, \quad \psi_n \propto e^{in\phi}$$

**Spectrum:**

$$E_n = \frac{\hbar^2}{2mR^2} \left( n - \frac{q\Phi_B}{\hbar c} \right)^2$$

(For uniform field  $B$  inside ring:  $\Phi_B = \pi R^2 B$ ).

- **Flux Quantum:**  $\Phi_0 = \hbar c/q$ . If  $\Phi_B = k\Phi_0$  ( $k \in \mathbb{Z}$ ), spectrum is unchanged.
- **Persistent Current:**  $J = -c \frac{\partial E_n}{\partial \Phi_B}$  (non-zero even in ground state).

**Mechanical Momentum:**  $\Pi = \mathbf{p} - \frac{q}{c} \mathbf{A}$ . Commutator  $[\Pi_i, \Pi_j] = i\hbar \frac{q}{c} \epsilon_{ijk} B_k$ .

**Drift Velocity:**  $\mathbf{v}_D = c \frac{\mathbf{E} \times \mathbf{B}}{B^2}$  (independent of  $q, m$ ).

$\nabla \times \mathbf{A}(\mathbf{r}) = \mathbf{B}(\mathbf{r})$ .

### 10.4 Zeeman Effect

$$\hat{H}' = -\frac{e}{2m_e c} (\hat{\mathbf{L}} + 2\hat{\mathbf{S}}) \cdot \mathbf{B}.$$

**Normal Zeeman (spin ignored):**  $\Delta E = m_\ell \mu_B B$ .

**Anomalous Zeeman (with spin):**

$$\Delta E = g_J m_J \mu_B B, \quad g_J = 1 + \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)}$$

**Paschen-Back (strong field):**  $L$ - $S$  coupling broken;  $\Delta E = (m_\ell + 2m_s) \mu_B B$ .

**Diamagnetic correction** (ground-state H,  $1s$ ):  $E_{\text{dia}}^{(1)} = \frac{e^2 B^2}{4m_e} \langle r^2 \rangle_{1s} = \frac{e^2 B^2 a_0^2}{4m_e Z^2}$ .

**Quadratic Stark** on H ( $n=2$ ,  $V = \lambda(x^2 - y^2)$ ): Only  $|2, 1, \pm 1\rangle$  mix;  $E^{(1)} = \pm 12\lambda a_0^2$ ,  $|\psi_\pm\rangle = \frac{1}{\sqrt{2}}(|2, 1, 1\rangle \mp |2, 1, -1\rangle)$ .  $2s$  and  $2p_0$  get zero first-order correction.

## 11 Perturbation Theory

**Validity condition:**  $\left| \frac{\langle k^{(0)} | H' | n^{(0)} \rangle}{E_n^{(0)} - E_k^{(0)}} \right| \ll 1$

### 11.1 Time-Independent — Non-Degenerate

$$\hat{H} = \hat{H}^{(0)} + \lambda V. \quad \hat{H}^{(0)} |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle.$$

Expand the exact eigenstates and eigenvalues in powers of  $\lambda$ :

$$|n\rangle = \sum_{k=0}^{\infty} \lambda^k |n^{(k)}\rangle, \quad E_n = \sum_{k=0}^{\infty} \lambda^k E_n^{(k)}$$

The perturbation equation:

$$(\hat{H}_0 + \lambda H') \left( \sum_{k=0}^{\infty} \lambda^k |n^{(k)}\rangle \right) = \left( \sum_{k=0}^{\infty} \lambda^k E_n^{(k)} \right) \left( \sum_{j=0}^{\infty} \lambda^j |n^{(j)}\rangle \right)$$

**First order:**

$$E_n^{(1)} = \langle n^{(0)} | V | n^{(0)} \rangle, \quad |n^{(1)}\rangle = \sum_{k \neq n} \frac{\langle k^{(0)} | V | n^{(0)} \rangle}{E_n^{(0)} - E_k^{(0)}} |k^{(0)}\rangle$$

**Second order (energy):**

$$E_n^{(2)} = \sum_{k \neq n} \frac{|\langle k^{(0)} | V | n^{(0)} \rangle|^2}{E_n^{(0)} - E_k^{(0)}}$$

**Substitution:** Identify which  $k$  states produce a non-zero integral. These are the only terms you keep in the sum.

$$V_{kn} = \langle k^{(0)} | H' | n^{(0)} \rangle = \int \psi_k^{(0)*}(x) H'(x) \psi_n^{(0)}(x) dx$$

In most physics problems, this integral is zero for many values of  $k$  due to symmetry (parity, orthogonality, angular momentum conservation).

**Closure approximation** (when excitation energies  $\sim \overline{\Delta E}$ ):

$$E_n^{(2)} \approx \frac{\langle n^{(0)} | V^2 | n^{(0)} \rangle - |V_{nn}|^2}{\overline{\Delta E}} = \frac{\sigma_V^2}{\overline{\Delta E}}$$

**Variance sum rule:**  $\sum_{k \neq n} |V_{kn}|^2 = \langle n^{(0)} | V^2 | n^{(0)} \rangle - |\langle n^{(0)} | V | n^{(0)} \rangle|^2 \equiv \sigma_V^2$

(useful sanity check; follows from completeness  $\sum_k |k\rangle \langle k| = \mathbb{I}$ ).

**Example:** For harmonic oscillator ground state with  $V = \lambda x^4$ ,  $E_0^{(1)} = \frac{3\lambda \hbar^2}{4m^2 \omega^2}$ .

### 11.2 Time-Independent — Degenerate

If  $E_n^{(0)}$  is  $g$ -fold degenerate with states  $\{|n_\alpha^{(0)}\rangle\}_{\alpha=1}^g$ , diagonalize the  $g \times g$  matrix:

$$W_{\alpha\beta} = \langle n_\alpha^{(0)} | V | n_\beta^{(0)} \rangle$$

Eigenvalues of  $W$  give the first-order corrections  $E_n^{(1)}$ ; solve the secular equation  $\det(W - E^{(1)} I) = 0$ .

**Second-order degenerate correction**

If degeneracy persists after first order ( $W_{\alpha\alpha} = W_{\beta\beta}$ ), define the effective second-order matrix in the still-degenerate subspace:

$$M_{\alpha\beta} = \sum_{k \notin D} \frac{\langle n_\alpha^{(0)} | V | k^{(0)} \rangle \langle k^{(0)} | V | n_\beta^{(0)} \rangle}{E_n^{(0)} - E_k^{(0)}}$$

Diagonalise  $M$  to lift remaining degeneracy; eigenvalues give  $E_n^{(2)}$ . *Hierarchical*: repeat until degeneracy is fully removed.

**Wave-function correction (outside degenerate subspace):**

$$\langle k^{(0)} | n, s^{(1)} \rangle = \frac{\langle k^{(0)} | V | n, s^{(0)} \rangle}{E_n^{(0)} - E_k^{(0)}} \quad (k \notin D)$$

### 11.3 Time-Dependent Perturbation Theory

$\hat{H} = \hat{H}^{(0)} + V(t)$ , with  $V(t)$  turned on at  $t = 0$ . Transition amplitude:

$$c_{i \rightarrow f}^{(1)}(t) = \frac{-i}{\hbar} \int_0^t \langle f | V(t') | i \rangle e^{i\omega_{fi} t'} dt', \quad \omega_{fi} = \frac{E_f - E_i}{\hbar}$$

$$P_{n \rightarrow m}^{(1)}(t) = |c_m^{(1)}(t)|^2 = \frac{1}{\hbar^2} \left| \int_0^t \langle m | V(t') | n \rangle e^{i\omega_{mn} t'} dt' \right|^2$$

**Constant perturbation:**

$$P_{i \rightarrow f}(t) = \frac{2|V_{fi}|^2}{\hbar^2} \frac{1 - \cos(\omega_{fi} t)}{\omega_{fi}^2} = \frac{|V_{fi}|^2}{\hbar^2} t^2 \text{sinc}^2\left(\frac{\omega_{fi} t}{2}\right)$$

**Sudden & adiabatic limits:** Pulse with decay  $e^{-t/\tau}$ :  $P_{fi} \propto |V_{fi}|^2 / (\hbar^2(\omega_{fi}^2 + 1/\tau^2))$ .  $\tau \rightarrow 0$  (sudden):  $P \propto \tau^2 \rightarrow 0$ ;  $\tau \rightarrow \infty$  (adiabatic):  $P \rightarrow |V_{fi}|^2 / (\hbar \omega_{fi})^2$ .

**HO impulse** ( $V = F_0 \hat{x}$  for  $0 < t < T$ ): with  $\lambda = \frac{F_0}{\sqrt{2m\hbar\omega^3}}$ :  $P_{0 \rightarrow 1} = 4\lambda^2 \sin^2 \frac{\omega T}{2}$ ,  $P_{0 \rightarrow 2} = 8\lambda^4 \sin^4 \frac{\omega T}{2}$ .

**Exact Dynamics**

Let  $H(t) = H_0 + V(t)$ . The state is  $|\psi(t)\rangle = \sum_n c_n(t) e^{-iE_n t/\hbar} |n\rangle$ . Exact coupled equations:

$$\dot{c}_n(t) = -\frac{i}{\hbar} \sum_m e^{i\omega_{nm} t} V_{nm}(t) c_m(t), \quad \omega_{nm} \equiv \frac{E_n - E_m}{\hbar}$$

**Second-order Dyson** ( $V$  small, system starts in  $|i\rangle$  at  $t \rightarrow -\infty$ ):

$$c_f^{(2)}(t) = -\frac{1}{\hbar^2} \sum_m \int_{-\infty}^t dt' \int_{-\infty}^{t'} dt'' e^{i\omega_{fm} t'} V_{fm}(t') e^{i\omega_{mi} t''} V_{mi}(t'')$$

**Quantum Quench** (smooth ramp  $V(-\infty) = 0$ ):

$$P(i \rightarrow f) = \frac{1}{(E_f - E_i)^2} \left| \int_{-\infty}^{\infty} dt' \dot{V}_{fi}(t') e^{i\omega_{fi} t'} \right|^2$$

## 11.4 Fermi's Golden Rule

**Periodic perturbation**  $V(t) = U e^{-i\omega t} + U^\dagger e^{i\omega t}$ :

- **Absorption** ( $E_f = E_i + \hbar\omega$ ):  $\Gamma = \frac{2\pi}{\hbar} |U_{fi}|^2 \delta(E_f - E_i - \hbar\omega)$ .
- **Stimulated emission** ( $E_f = E_i - \hbar\omega$ ):  $\Gamma = \frac{2\pi}{\hbar} |U_{if}|^2 \delta(E_f - E_i + \hbar\omega)$ .

With a continuum of final states (density  $\rho$ ):

$$\Gamma_{i \rightarrow f} = \frac{2\pi}{\hbar} |U_{fi}|^2 \rho(E_f)$$

The density of states  $\rho(E) = \frac{L^3}{2\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \sqrt{E}$ .

**Level width & lifetime:**  $P_i(t) = e^{-\Gamma t}$ ,  $\tau = 1/\Gamma$ , natural linewidth  $\Delta E \approx \hbar\Gamma$ .  
Formal:  $E_i \rightarrow E_i - i\hbar\Gamma/2$  gives Lorentzian lineshape.

## 11.5 Scattering and Born Approximation

In scattering theory, the T-matrix operator satisfies the Lippmann-Schwinger equation:

$$T = V + V \frac{1}{E - H_0 + i\epsilon} T$$

The scattering amplitude is:

$$f(\theta) = -\frac{m}{2\pi\hbar^2} \langle \mathbf{k}_f | T | \mathbf{k}_i \rangle$$

In the Born approximation (first order),  $T \approx V$

## 12 Identical Particles

**Symmetrization postulate:** The state of  $N$  identical particles must be

- **Symmetric** under exchange  $\Rightarrow$  **Bosons** (integer spin).
- **Antisymmetric** under exchange  $\Rightarrow$  **Fermions** (half-integer spin).

**Two particles:**

$$|\psi\rangle_{S/A} = \frac{1}{\sqrt{2}} (|a\rangle |b\rangle \pm |b\rangle |a\rangle)$$

### 12.1 Derivation: Pauli Exclusion Principle

Indistinguishability requires  $|\Psi|^2$  invariance under particle exchange. Define exchange operator  $\hat{P}_{ij}$ :  $\hat{P}_{ij}\Psi = \lambda\Psi$ . Since  $\hat{P}^2 = \mathbb{I} \Rightarrow \lambda = \pm 1$ . Fermions are defined by the antisymmetric sector ( $\lambda = -1$ ):  $\Psi(x_j, x_i) = -\Psi(x_i, x_j)$ .

$$\Psi(x_1, \dots, x_N) = \frac{1}{\sqrt{N!}} \det \begin{pmatrix} \phi_{a_1}(x_1) & \dots & \phi_{a_1}(x_N) \\ \vdots & \ddots & \vdots \\ \phi_{a_N}(x_1) & \dots & \phi_{a_N}(x_N) \end{pmatrix} \quad (1)$$

Exchange of particles  $i \leftrightarrow j$  corresponds to swapping columns  $i, j$ , introducing a factor of  $(-1)$ , satisfying the fermion condition.

**3. Interpretation (Coincidence Limit):** Assume two fermions occupy the same state  $\alpha_k = \alpha_m$  (for  $k \neq m$ ). This implies Row  $k$  and Row  $m$  are identical:

$$\phi_{\alpha_k}(x) = \phi_{\alpha_m}(x) \quad \forall x \Rightarrow \det(\mathbf{M}) = 0$$

**Conclusion:**  $\Psi \equiv 0$  everywhere. The probability density vanishes, proving two fermions cannot occupy the same quantum state.

**Exchange interaction:** For two electrons in states  $a, b$ :

$$E = E_{\text{direct}} \pm E_{\text{exchange}}$$

$$E_{\text{ex}} = \int \phi_a^*(\mathbf{r}_1) \phi_b^*(\mathbf{r}_2) V(\mathbf{r}_1, \mathbf{r}_2) \phi_b(\mathbf{r}_1) \phi_a(\mathbf{r}_2) d^3r_1 d^3r_2$$

**Antisymmetry constraint** (two identical fermions):  $(-1)^L \cdot (-1)^{S+1} = -1 \Leftrightarrow L+S$  even.

E.g.  $S = 0 \Rightarrow$  even  $L$ ;  $S = 1 \Rightarrow$  odd  $L$ .

**Addition of three spins**  $\frac{1}{2} \oplus \frac{1}{2} \oplus \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2} \oplus \frac{1}{2}$  (dim  $4 + 2 + 2 = 8$ ).

**Spin-orbit time evolution** ( $H = \frac{\alpha}{\hbar} \vec{L} \cdot \vec{S}$ ,  $\ell = 1, s = \frac{1}{2}$ ):

$$E_{j=3/2} = \frac{\alpha\hbar}{2}, \quad E_{j=1/2} = -\alpha\hbar.$$

Transition probability ( $m_j = -\frac{1}{2}$  subspace):  $P_{(0, \downarrow) \rightarrow (-1, \uparrow)}(t) = \frac{8}{9} \sin^2\left(\frac{3\alpha t}{4}\right)$ .

## 13 Variational Methods — Theorems & Proofs

### 13.1 The Variational Theorem

**Theorem.** For any normalizable trial state  $|\psi\rangle$  in the domain of  $\hat{H}$ :

$$E[\psi] \equiv \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} \geq E_0$$

**Assumptions.**  $\hat{H}$  is Hermitian with discrete spectrum bounded below:  $E_0 \leq E_1 \leq \dots$ . The eigenstates  $\{|n\rangle\}$  form a complete orthonormal basis:  $\hat{H}|n\rangle = E_n|n\rangle$ ,  $\langle m|n\rangle = \delta_{mn}$ .

**Proof.** Expand the trial state in the *exact* eigenbasis to convert the expectation value into a weighted average of eigenvalues, then bound from below.

*Step 1 — Expand:*  $|\psi\rangle = \sum_n c_n |n\rangle$ , with  $c_n = \langle n | \psi \rangle$ . By orthonormality:

$$\langle \psi | \psi \rangle = \sum_n |c_n|^2,$$

$$\langle \psi | \hat{H} | \psi \rangle = \sum_{n,m} c_m^* c_n \langle m | \hat{H} | n \rangle = \sum_n |c_n|^2 E_n.$$

*Step 2 — Bound:* Since every eigenvalue satisfies  $E_n \geq E_0$ , replace  $E_n \rightarrow E_0$  in the numerator:

$$E[\psi] = \frac{\sum_n |c_n|^2 E_n}{\sum_n |c_n|^2} \geq \frac{E_0 \sum_n |c_n|^2}{\sum_n |c_n|^2} = E_0.$$

Equality iff  $c_n = 0 \forall n \neq 0$ , i.e.  $|\psi\rangle \propto |0\rangle$  (assuming  $E_0$  non-degenerate). ■

**Stationarity (Generalized Ritz).**  $\delta E[\psi] = 0 \Leftrightarrow \hat{H}|\psi\rangle = E|\psi\rangle$ .

*Proof sketch:* Write  $E \langle \psi | \psi \rangle = \langle \psi | \hat{H} | \psi \rangle$ ; vary  $\langle \psi | \rightarrow \langle \psi | + \langle \delta \psi |$  independently  $\Rightarrow (\hat{H} - E)|\psi\rangle = 0$ .

### 13.2 The Ritz Linear Method (Ritz–Galerkin)

**Ansatz.** Restrict the trial state to an  $N$ -dim subspace  $\{|\phi_i\rangle\}_{i=1}^N$  (linearly independent, *not* necessarily orthogonal):

$$|\psi\rangle = \sum_{i=1}^N c_i |\phi_i\rangle.$$

Define the Hamiltonian and overlap matrices:  $H_{ij} = \langle \phi_i | \hat{H} | \phi_j \rangle$ ,  $S_{ij} = \langle \phi_i | \phi_j \rangle$ .

**Derivation.** enforce  $\partial E / \partial c_k^* = 0$  to reduce the variational condition to a matrix eigenvalue problem.

*Step 1 — Rewrite the functional:*  $E[\mathbf{c}] = \mathbf{c}^\dagger \mathbf{H} \mathbf{c} / \mathbf{c}^\dagger \mathbf{S} \mathbf{c}$ , equivalently

$$\sum_{ij} c_i^* (H_{ij} - E S_{ij}) c_j = 0.$$

*Step 2 — Differentiate* w.r.t. each  $c_k^*$  (treating  $c_k$  and  $c_k^*$  as independent):

$$\frac{\partial}{\partial c_k^*} \sum_{ij} c_i^* (H_{ij} - E S_{ij}) c_j = \sum_j (H_{kj} - E S_{kj}) c_j = 0, \quad k = 1, \dots, N.$$

*Step 3 — Matrix form:* This is the generalized eigenvalue problem  $\mathbf{H} \mathbf{c} = E \mathbf{S} \mathbf{c}$ . Non-trivial  $\mathbf{c} \neq \mathbf{0}$  exists iff:

$$\det(\mathbf{H} - E \mathbf{S}) = 0 \quad (\text{Secular equation})$$

The  $N$  roots  $E^{(1)} \leq \dots \leq E^{(N)}$  are upper bounds to the first  $N$  exact eigenvalues:  $E^{(k)} \geq E_k^{\text{exact}}$  (interleaving theorem).

### 13.3 The Variance Theorem

**Theorem.** For any  $|\psi\rangle$  with  $E = \langle \psi | \hat{H} | \psi \rangle$  and variance  $\sigma^2 = \langle \psi | \hat{H}^2 | \psi \rangle - E^2$ , at least one exact eigenvalue  $E_n$  satisfies:

$$|E - E_n| \leq \sigma$$

**Proof** (by contradiction). *Key idea:* if every eigenvalue were far from  $E$ , the variance sum would exceed itself.

*Step 1 — Expand variance:* Using  $|\psi\rangle = \sum_n c_n |n\rangle$  and orthonormality:

$$\sigma^2 = \langle \psi | (\hat{H} - E)^2 | \psi \rangle = \sum_n |c_n|^2 (E_n - E)^2.$$

*Step 2 — Assume the contrary:* Suppose  $|E_n - E| > \sigma$  for all  $n$ . Then  $(E_n - E)^2 > \sigma^2$  for every term, so

$$\sigma^2 = \sum_n |c_n|^2 (E_n - E)^2 > \sigma^2 \sum_n |c_n|^2 = \sigma^2.$$

*Step 3:*  $\sigma^2 > \sigma^2$  is a contradiction. Hence  $\exists E_n$  with  $|E - E_n| \leq \sigma$ . ■

*Physical use:* a small variance  $\sigma$  guarantees the trial energy  $E$  is close to a true eigenvalue; combines with the variational bound to bracket  $E_0$ .

### 13.4 The Min-Max Theorem (Courant–Fischer)

Characterizes  $E_k$  without requiring explicit orthogonality to lower states.

$$E_k = \min_{\dim \mathcal{W}=k+1} \left( \max_{|\psi\rangle \in \mathcal{W} \setminus \{0\}} \frac{\langle \psi | \hat{H} | \psi \rangle}{\langle \psi | \psi \rangle} \right)$$

**Proof sketch.** Key idea: any  $(k+1)$ -dim subspace must “touch” the  $k$ -th eigenstate by a dimension-counting argument.

**Step 1 — Lower bound:** Let  $\mathcal{W}$  be any  $(k+1)$ -dim subspace. The space orthogonal to  $\{|0\rangle, \dots, |k-1\rangle\}$  has codimension  $k$ , so  $\mathcal{W}$  intersects it non-trivially:  $\exists |\phi\rangle \in \mathcal{W}$  with  $\langle n | \phi \rangle = 0$  for  $n < k$ .

**Step 2:** Expand  $|\phi\rangle = \sum_{n \geq k} c_n |n\rangle$ . Since  $E_n \geq E_k$  for all  $n \geq k$ :

$$\frac{\langle \phi | \hat{H} | \phi \rangle}{\langle \phi | \phi \rangle} = \frac{\sum_{n \geq k} |c_n|^2 E_n}{\sum_{n \geq k} |c_n|^2} \geq E_k.$$

Because  $|\phi\rangle \in \mathcal{W}$ , the maximum Rayleigh quotient over  $\mathcal{W}$  is  $\geq E_k$ .

**Step 3 — Achievability:** Choose  $\mathcal{W}^* = \text{span}\{|0\rangle, \dots, |k\rangle\}$ . The highest eigenvalue in  $\mathcal{W}^*$  is exactly  $E_k$ , so the minimum over all subspaces equals  $E_k$ . ■

## 14 Quantum Information Basics

A **qubit** is a two-level quantum system, represented as  $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$ , with  $|\alpha|^2 + |\beta|^2 = 1$ .

The **Bloch sphere** visualizes the qubit state:  $|\psi\rangle = \cos(\theta/2) |0\rangle + e^{i\phi} \sin(\theta/2) |1\rangle$ , with  $\mathbf{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ .

**Basic gates:** Pauli matrices as rotations:  $X = \sigma_x$  (bit flip),  $Y = \sigma_y$ ,  $Z = \sigma_z$  (phase flip),  $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  (Hadamard).

## 15 Fine Structure of Hydrogen

$$\hat{H} = \hat{H}_0 + H_{\text{FS}}, \text{ where } \hat{H}_0 = \frac{p^2}{2m} - \frac{e^2}{4\pi\epsilon_0 r}.$$

$$H_{\text{FS}} = H_{\text{rel}} + H_{\text{SO}} + H_{\text{Darwin}}$$

### 15.1 The Three Corrections

#### 1. Relativistic kinetic:

$$H_{\text{rel}} = -\frac{p^4}{8m^3 c^2}$$

(from Taylor-expanding  $E = \sqrt{p^2 c^2 + m^2 c^4}$ ).

#### 2. Spin–orbit coupling:

$$H_{\text{SO}} = \frac{e^2}{8\pi\epsilon_0 m^2 c^2 r^3} \vec{L} \cdot \vec{S}$$

(includes Thomas precession factor  $\frac{1}{2}$ ).

Use  $\vec{L} \cdot \vec{S} = \frac{1}{2}(\hat{J}^2 - \hat{L}^2 - \hat{S}^2)$  to diagonalise in the **coupled basis**  $|n, \ell, s, j, m_j\rangle$ .

#### 3. Darwin term (affects only $\ell = 0$ ):

$$H_{\text{Darwin}} = \frac{\hbar^2}{8m^2 c^2} 4\pi \frac{e^2}{4\pi\epsilon_0} \delta^{(3)}(\mathbf{r})$$

All three are  $\mathcal{O}(\alpha^2 E_n^{(0)}) \sim \alpha^4 m c^2 / n^3$ .

### 15.2 Energy Correction

$$E_{nj}^{(1)} = \frac{E_n^{(0)} \alpha^2}{n^2} \left( \frac{n}{j + \frac{1}{2}} - \frac{3}{4} \right)$$

or equivalently

$$\Delta E_{\text{FS}} = -\frac{m c^2 \alpha^4}{2 n^3} \left( \frac{1}{j + \frac{1}{2}} - \frac{3}{4 n} \right)$$

Depends on  $n$  and  $j$  only (not  $\ell$  alone). States with same  $j$  but different  $\ell$  remain degenerate (e.g.  $2s_{1/2}$  and  $2p_{1/2}$ ).

**Good quantum numbers:**  $n, \ell, s, j, m_j$ .

## 16 Scattering Theory

### 16.1 Setup & Cross-Section

Scattering state:  $\psi(\mathbf{r}) \xrightarrow{r \rightarrow \infty} e^{ikz} + f(\theta, \phi) \frac{e^{ikr}}{r}$ .

**Differential cross-section:**

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2$$

**Total:**  $\sigma = \int |f(\theta)|^2 d\Omega$ .

### 16.2 Born Approximation (1st order)

$$f(\theta) = -\frac{m}{2\pi\hbar^2} \int V(\mathbf{r}') e^{i\mathbf{q} \cdot \mathbf{r}'} d^3 r'$$

momentum transfer:  $\mathbf{q} = \mathbf{k}_i - \mathbf{k}_f$ ,  $|\mathbf{q}| = 2k \sin(\theta/2)$ ,  $k = \sqrt{2\mu E}/\hbar = \frac{p}{\hbar}$ .

**Note:**  $q = |\mathbf{q}|$  is the momentum transfer;  $k = |\mathbf{k}_i|$  is the incident wave number. Do not confuse them in the Born formula.

**Validity condition** ( $V$  must be a weak perturbation on the free wave):

$$\frac{m}{\hbar^2} \left| \int \frac{e^{ikr}}{r} V(r) d^3 r \right| \ll 1$$

For Yukawa  $V = V_0 e^{-\mu r}/r$  this gives two limits:

$$\text{Low energy } (k\mu^{-1} \ll 1) : \quad \frac{2mV_0}{\hbar^2 \mu^2} \ll 1$$

$$\text{High energy } (k\mu^{-1} \gg 1) : \quad \frac{2mV_0}{\hbar^2 \mu k} \ll 1$$

Born improves at high energy; always breaks down for strong/long-range potentials.

For a **spherically symmetric** potential:

$$f(\theta) = -\frac{2m}{\hbar^2 q} \int_0^\infty r V(r) \sin(qr) dr$$

**Born for spherical well** ( $V = -V_0$  for  $r \leq R$ , else 0):

$$f(\theta) = \frac{2mV_0}{\hbar^2 q^3} [\sin(qR) - qR \cos(qR)]$$

**Form factor:** Extended charge  $\rho(r)$  vs. point charge:  $f = F(q^2) f^{\text{point}}$ ,  $F(q^2) = 1 - \frac{q^2 \langle r^2 \rangle}{6} + \dots$

$$R_{\text{RMS}} = \sqrt{-6 dF/dq^2|_{q=0}}.$$

**3D delta**  $V = \lambda \delta^{(3)}(\mathbf{r})$ : no scattering for  $\ell \geq 1$  since  $R_\ell(0) = 0$ . If the scattering matrix is independent of the angle than  $\sigma_t = \int \frac{d\sigma}{d\Omega} d\Omega = \int |f|^2 d\Omega = 4\pi |f|^2$

### 16.3 Yukawa Potential Scattering

Yukawa potential:  $V(r) = -\frac{g^2}{r} e^{-r/a}$ ;  $q = 2k \sin(\theta/2)$ .

**Born approximation:**

$$f(\theta) = -\frac{2mg^2/\hbar^2}{q^2 + 1/a^2}, \quad q = 2k \sin(\theta/2)$$

Signs: attractive potential ( $g^2 > 0$ ,  $V < 0$ ) gives  $f > 0$  in the convention  $V = -g^2/r$  with the  $-$  already absorbed. **Coulomb limit** ( $a \rightarrow \infty$ ,  $\mu = 1/a \rightarrow 0$ ):

$$f(\theta) = -\frac{2mg^2}{\hbar^2 q^2}, \quad \frac{d\sigma}{d\Omega} = \left( \frac{2mg^2}{\hbar^2 q^2} \right)^2$$

### 16.4 Photoabsorption Cross-Section

For EM radiation (dipole approx.  $e^{i\mathbf{k} \cdot \mathbf{r}} \approx 1$ ):

$$\sigma = \frac{\Gamma}{F}, \quad F = \frac{2c\epsilon_0 \omega A_0^2}{\hbar}$$

Using Fermi's Golden Rule:

$$\sigma_{i \rightarrow f} = \frac{4\pi^2 \alpha \hbar}{m^2 \omega} |\langle f | \epsilon \cdot \hat{\mathbf{p}} | i \rangle|^2 \rho(E_f)$$



## 16.5 Lippmann-Schwinger Equation

The Lippmann-Schwinger equation for the scattering wave function:

$$\psi^{(+)}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} + \int d^3r' G_0^{(+)}(\mathbf{r}, \mathbf{r}') V(\mathbf{r}') \psi^{(+)}(\mathbf{r}')$$

Where the free Green's function is  $G_0^{(+)}(\mathbf{r}, \mathbf{r}') = -\frac{m}{2\pi\hbar^2} \frac{e^{ik|\mathbf{r}-\mathbf{r}'|}}{|\mathbf{r}-\mathbf{r}'|}$ .

The Born approximation is the first iteration:  $\psi^{(+)} \approx e^{i\mathbf{k}\cdot\mathbf{r}} + \int G_0^{(+)} V e^{i\mathbf{k}\cdot\mathbf{r}'}$ .

The T-matrix satisfies  $T = V + V G_0^{(+)} T$ , and the scattering amplitude is  $f(\theta) = -\frac{m}{2\pi\hbar^2} \langle \mathbf{k}' | T | \mathbf{k} \rangle$ .

For high-energy scattering, the eikonal approximation approximates the phase shift as  $\chi(b) = -\frac{m}{2\pi\hbar^2 k} \int_{-\infty}^{\infty} V(\sqrt{b^2 + z^2}) dz$ .

## 17 Finite Potential Well (Bound States)

$$V = \begin{cases} -U_0 & |x| \leq L/2 \\ 0 & \text{otherwise} \end{cases}; \quad k = \sqrt{2m(E + U_0)/\hbar^2}, \quad \kappa = \sqrt{-2mE/\hbar^2}, \quad k_0^2 = k^2 + \kappa^2.$$

**Even (symmetric):**  $\psi \propto \cos(kx)$  inside;  $e^{-\kappa|x|}$  outside.

$$\kappa = k \tan\left(\frac{kL}{2}\right) \Leftrightarrow \sqrt{(k_0/k)^2 - 1} = \tan\left(\frac{kL}{2}\right)$$

**Odd (antisymmetric):**  $\psi \propto \sin(kx)$  inside;  $\text{sgn}(x)e^{-\kappa|x|}$  outside.

$$\kappa = -k \cot\left(\frac{kL}{2}\right) \Leftrightarrow \sqrt{(k_0/k)^2 - 1} = -\cot\left(\frac{kL}{2}\right)$$

Solve graphically: at least one bound state always exists.

## 18 Wave Packet Dynamics

### 18.1 Free Gaussian Wave Packet

Initial:  $\psi(x, 0) = (2\pi\sigma_0^2)^{-1/4} e^{ip_0x/\hbar} e^{-x^2/4\sigma_0^2}$ .

Width at time  $t$ :

$$\sigma(t) = \sigma_0 \sqrt{1 + \left(\frac{\hbar t}{2m\sigma_0^2}\right)^2}$$

Disperses:  $\sigma \rightarrow \infty$  as  $t \rightarrow \infty$ .  $|\psi(x, t)|^2$  remains Gaussian centred at  $x = v_0 t$  ( $v_0 = p_0/m$ ).

**Momentum-space:**  $\tilde{\psi}(p, t) = \tilde{\psi}(p, 0) e^{-ip^2 t/(2m\hbar)}$ ;  $\tilde{\psi}(p, 0) = (2\sigma_0^2/\pi\hbar^2)^{1/4} e^{-\sigma_0^2(p-p_0)^2/\hbar^2}$ .  $\langle \hat{p} \rangle = p_0$ ,  $\sigma_p = \hbar/(2\sigma_0)$  (constant in time);  $\sigma_x \sigma_p = \hbar/2$  only at  $t = 0$ .

### 18.2 Probability Current via Phase

Write  $\psi = \sqrt{\rho} e^{iS/\hbar}$ . Then:

$$\mathbf{j} = \frac{\rho \nabla S}{m}, \quad \mathbf{v} = \frac{\nabla S}{m}$$

Plane wave  $\psi \propto e^{ipx/\hbar}$ :  $S = px$ ,  $v = p/m$ .

## 19 Isotropic Harmonic Oscillators (2D & 3D)

### 19.1 3D Isotropic HO

$$\hat{H} = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 r^2. \quad [\hat{H}, \hat{L}_i] = 0, \text{ so } [\hat{H}, \hat{L}^2] = [\hat{H}, \hat{L}_z] = 0.$$

Eigenstates  $|n_r, \ell, m\rangle$  ( $n_r$  = radial quantum number):

$$E = \hbar\omega\left(2n_r + \ell + \frac{3}{2}\right), \quad n_r \geq 0, \ell \geq 0$$

With  $N = 2n_r + \ell$ :  $E_N = \hbar\omega(N + \frac{3}{2})$ . Degeneracy:  $g_N = \frac{(N+1)(N+2)}{2}$ .

### 19.2 2D Isotropic HO

$$\hat{H} = \hbar\omega(\hat{a}_x^\dagger \hat{a}_x + \hat{a}_y^\dagger \hat{a}_y + 1).$$

**Circular ladder operators:**

$$\hat{a}_R = \frac{\hat{a}_x - i\hat{a}_y}{\sqrt{2}}, \quad \hat{a}_L = \frac{\hat{a}_x + i\hat{a}_y}{\sqrt{2}}$$

$[\hat{a}_L, \hat{a}_L^\dagger] = [\hat{a}_R, \hat{a}_R^\dagger] = 1$ ; cross-commutators vanish.

$$\hat{H} = \hbar\omega(\hat{n}_R + \hat{n}_L + 1), \quad \hat{L}_z = \hbar(\hat{n}_R - \hat{n}_L)$$

$[\hat{H}, \hat{L}_z] = 0$ ; joint eigenstates  $|n_R, n_L\rangle$ .

Energy:  $E_N = \hbar\omega(N + 1)$  with  $N = n_R + n_L$ . Degeneracy:  $g_N = N + 1$ .

## 20 Coupled Oscillators

Two particles coupled by spring  $\lambda$ :

$$\hat{H} = \frac{p_1^2 + p_2^2}{2m} + \frac{m\omega^2}{2} [(x_1 - a)^2 + (x_2 + a)^2] + \lambda m \omega^2 (x_1 - x_2)^2$$

**Normal-mode coordinates:**  $x_{\text{cm}} = \frac{x_1 + x_2}{2}$ ,  $x_R = x_1 - x_2$ ;  $p_{\text{cm}} = p_1 + p_2$ ,  $p_R = \frac{p_1 - p_2}{2}$ .

Decouples:  $\hat{H} = \hat{H}_{\text{cm}} + \hat{H}_R + C$ ;  $\omega_{\text{cm}} = \omega$ ,  $\omega_R = \omega\sqrt{1 + 4\lambda}$ .

Effective masses:  $2m$  (cm),  $m/2$  (relative).

$$E_{n_{\text{cm}}, n_R} = \hbar\omega\left(n_{\text{cm}} + \frac{1}{2}\right) + \hbar\omega_R\left(n_R + \frac{1}{2}\right) + C$$

$C = m\omega^2 a^2 \frac{4\lambda}{4\lambda + 1}$ . Wave function:  $\psi(x_{\text{cm}}, x_R) = \psi_{\text{cm}}(x_{\text{cm}}) \psi_R(x_R)$ .

## 21 Noether's Theorem & Symmetry-Conservation

Infinitesimal change:  $\delta\hat{q}_i = -i\varepsilon[\hat{Q}, \hat{q}_i] = \varepsilon F_i(\hat{\mathbf{q}})$ .

**Symmetry  $\Leftrightarrow$  conservation:**  $[\hat{H}, \hat{X}] = 0 \Leftrightarrow \hat{X}$  conserved  $\Leftrightarrow e^{i\theta\hat{X}}$  is a symmetry.

Infinitesimal operator transformation:  $\delta\hat{O} = i\varepsilon[\hat{X}, \hat{O}]$ .

### 21.1 Constraints on Attractive Potentials

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 - \frac{\alpha}{r^s}, \quad \alpha > 0:$$

- $s > 2$ : spectrum unbounded below (non-physical).
- $s < 2$ : infinitely many bound states. ( $s = 2$  is marginal.)

**Proof idea** ( $s > 2$ ): Try  $\Psi(\mathbf{r}) = N e^{-r^2/r_0^2}$ . Then  $\langle T \rangle \sim \hbar^2/(2mr_0^2)$ ,  $\langle V \rangle \sim -\alpha/r_0^s$ , so  $E(r_0) \approx \alpha_1/r_0^2 - \alpha_2/r_0^s \rightarrow -\infty$  as  $r_0 \rightarrow 0$ .

## 22 Group Theory & Representations

### 22.1 Definitions

A **representation**  $D$  of group  $G$  maps  $g \in G$  to operators  $D(g)$  on  $\mathcal{H}$  preserving group structure:

$$D(g_1)D(g_2) = D(g_1 g_2), \quad D(g^{-1}) = D(g)^{-1}$$

Unitary rep:  $D(g)^\dagger = D(g)^{-1}$ . **Reducible:** Can be decomposed into subspace actions. **Irreducible (Irrep):** No invariant subspaces (except  $\{0\}, \mathcal{H}$ ).

### 22.2 24.2 Schur's Lemmas

**Lemma 1:** If an operator  $\hat{A}$  commutes with all matrices of an irreducible representation  $D(g)$  (i.e.  $[\hat{A}, D(g)] = 0 \forall g$ ), then  $\hat{A}$  is a scalar multiple of the identity:

$$\hat{A} = \lambda \mathbb{I}$$

**Consequence:** In an irrep subspace, any constant of motion (symmetry) is trivial. *Proof (Hermitian case):*  $\hat{A}$  has an eigenvector  $v$  with  $\hat{A}v = \lambda v$ . The span of  $\{D(g)v\}$  is invariant and nonzero; by irreducibility it is all of  $\mathcal{H}$ . Thus  $\hat{A} = \lambda \mathbb{I}$  on the whole space.

**Lemma 2 (Orthogonality):** For two unitary irreps  $D^{(\mu)}$  and  $D^{(\nu)}$  of dimensions  $d_\mu, d_\nu$ :

$$\sum_{g \in G} D_{ij}^{(\mu)}(g)^* D_{k\ell}^{(\nu)}(g) = \frac{|G|}{d_\mu} \delta_{\mu\nu} \delta_{ik} \delta_{j\ell}$$

## 23 Solved Problem: Linear Rigid Rotor

### 23.1 Setup

A diatomic molecule with reduced mass  $\mu$  and fixed bond length  $r_0$ . The particle moves on the surface of a sphere of radius  $r_0$ .

$$\hat{H} = \frac{\hat{L}^2}{2I}, \quad I = \mu r_0^2$$

### 23.2 Exact Solution

Eigenstates are spherical harmonics  $Y_\ell^m(\theta, \phi)$ :

$$E_\ell = \frac{\hbar^2 \ell(\ell+1)}{2I}, \quad \ell = 0, 1, 2, \dots$$

Each level has degeneracy  $g_\ell = 2\ell + 1$  (from  $m = -\ell, \dots, \ell$ ).

**Rotational constant:**  $B = \frac{\hbar^2}{2I}$ ;  $E_\ell = B \ell(\ell+1)$ .

**Transition frequencies** (dipole selection rule  $\Delta\ell = \pm 1$ ):

$$\nu_{\ell \rightarrow \ell+1} = \frac{2B(\ell+1)}{h}, \quad \text{equally spaced by } 2B/h$$

### 23.3 Rigid Rotor in Electric Field (Perturbation)

$V = -\mathbf{d} \cdot \mathbf{E} = -dE \cos \theta$  where  $d$  is the dipole moment.

**First-order:**  $E_\ell^{(1)} = \langle \ell, m | (-dE \cos \theta) | \ell, m \rangle = 0$ . Zero because  $|Y_\ell^m|^2$  is even (parity +1),  $\cos \theta$  is odd (parity -1).

**Second-order** (using  $\cos \theta = \sqrt{\frac{4\pi}{3}} Y_1^0$ ): Sum covers  $J = \ell \pm 1$  ( $J = \ell$  forbidden by parity):

$$E_\ell^{(2)} = \sum_{J \neq \ell} \frac{| \langle J, m | (-dE \cos \theta) | \ell, m \rangle |^2}{E_\ell - E_J}$$

$$E_\ell^{(2)} = -\frac{d^2 E^2}{2B} \cdot \frac{\ell(\ell+1) - 3m^2}{(2\ell+3)(2\ell-1)\ell(\ell+1)}$$

For  $\ell = 0$ :  $E_0^{(2)} = -\frac{d^2 E^2}{6B}$ .

### 23.4 Particle on a Ring & Torsional Hindrance (Ethane)

**Unperturbed:**  $H_0 = L_z^2/2I$ ,  $I = ma^2$ ;  $\psi_n = e^{in\phi}/\sqrt{2\pi}$ ;  $E_n^{(0)} = \hbar^2 n^2/2I$ .  $n = 0$  non-degenerate;  $|n| \geq 1$  doubly degenerate ( $\pm n$ ).

**Key matrix element:**

$$\langle m | e^{ik\phi} | n \rangle = \delta_{m,n+k} \implies \langle m | \cos(k\phi) | n \rangle = \frac{1}{2}(\delta_{m,n+k} + \delta_{m,n-k})$$

**Selection rule:**  $H' = f(\phi)$  with Fourier mode  $e^{ik\phi} \Rightarrow \Delta n = \pm k$ .  $E_n^{(1)} = 0$  if  $f$  has no DC component ( $k \neq 0$ ). Denominator:  $E_n^{(0)} - E_m^{(0)} = \hbar^2(n^2 - m^2)/2I$ .

**Electric field**  $H' = -\mu \mathcal{E} \cos \phi$  ( $k = 1$ , gap  $= -\hbar^2/2I$ ):

$$|\Psi_0^{(1)}\rangle = |0\rangle + \frac{\mu \mathcal{E} I}{\hbar^2} (|1\rangle + |-1\rangle)$$

**Polarizability:**  $\langle \mu \cos \phi \rangle = \frac{2\mu^2 I}{\hbar^2} \mathcal{E} \implies \alpha = \frac{2\mu^2 I}{\hbar^2}$  ( $\alpha \propto I$ : larger  $I \Rightarrow$  smaller gap).

**Torsional hindrance**  $H' = W \cos(3\phi)$  ( $k = 3$ , gap  $= -9\hbar^2/2I$ ):

$$|\Psi_0^{(1)}\rangle = |0\rangle - \frac{WI}{9\hbar^2} (|3\rangle + |-3\rangle)$$

**Degenerate PT for  $n = \pm 3$ :** these states are coupled by  $H'$  since  $\langle +3 | \cos(3\phi) | -3 \rangle = \frac{1}{2}$ . Diagonalize  $\begin{pmatrix} 0 & W/2 \\ W/2 & 0 \end{pmatrix} \Rightarrow E_{\pm 3}^{(0)} \pm W/2$ ; eigenstates  $\propto \cos(3\phi), \sin(3\phi)$ .

## 24 Solved Problem: Stark Effect

External electric field  $\mathbf{E} = \mathcal{E} \hat{z}$ ; perturbation  $H' = e\mathcal{E}z = e\mathcal{E}r \cos \theta$ .

### 24.1 Ground State ( $n = 1$ ) — Quadratic Stark

$|1, 0, 0\rangle$  is non-degenerate. Parity  $\Rightarrow E_1^{(1)} = 0$ .

**Second-order:**

$$E_1^{(2)} = -\frac{9}{4} a_0^3 \mathcal{E}^2 \cdot \frac{4\pi\epsilon_0}{e^2} = -\frac{9}{4} \frac{a_0^3 \mathcal{E}^2}{E_1}$$

**Polarisability:**  $\alpha = \frac{9}{2}(4\pi\epsilon_0)a_0^3$ ;  $\Delta E = -\frac{1}{2}\alpha\mathcal{E}^2$ .

### 24.2 $n = 2$ — Linear Stark Effect

The  $n = 2$  level is 4-fold degenerate:  $|2, 0, 0\rangle, |2, 1, 0\rangle, |2, 1, \pm 1\rangle$ .

Parity ( $\ell \leftrightarrow \ell \pm 1$ ) and  $\Delta m = 0$  selection rules enforce zeros. Only  $s$ - $p$  mixing with  $m = 0$  survives:

$$\langle 2, 0, 0 | e\mathcal{E}z | 2, 1, 0 \rangle = \int_0^\infty R_{20} r R_{21} r^2 dr \int Y_0^0 \cos \theta Y_1^0 d\Omega = \frac{-3\sqrt{3}a_0}{\sqrt{3}}$$

Angular part:  $\cos \theta = \sqrt{4\pi/3} Y_1^0$ , so  $\int Y_0^0 \cos \theta Y_1^0 d\Omega = \sqrt{4\pi/3} \int |Y_1^0|^2 d\Omega / \sqrt{4\pi} = 1/\sqrt{3}$ . Radial part evaluated using  $\int_0^\infty r^n e^{-ar} dr = n!/\alpha^{n+1}$ . States  $|2, 1, \pm 1\rangle$  ( $m \neq 0$ ) have no partner to mix with ( $\Delta m = 0$ ).

The  $W$  matrix in the  $\{|2, 0, 0\rangle, |2, 1, 0\rangle\}$  subspace:

$$W = \begin{pmatrix} 0 & -3e\mathcal{E}a_0 \\ -3e\mathcal{E}a_0 & 0 \end{pmatrix}$$

$$E_{n=2}^{(1)} = \pm 3e\mathcal{E}a_0$$

Good states:  $|\pm\rangle = \frac{1}{\sqrt{2}}(|2, 0, 0\rangle \mp |2, 1, 0\rangle)$ .

### 24.3 27.3 General $n$ — Linear Stark

For level  $n$ , the first-order Stark splitting is:

$$\Delta E^{(1)} = \frac{3}{2} n(n_1 - n_2) e\mathcal{E}a_0$$

where  $n_1, n_2$  are parabolic quantum numbers ( $n_1 + n_2 + |m| + 1 = n$ ).

## 25 Solved Problem: Zeeman Effect

### 25.1 Weak-Field Zeeman ( $H_Z \ll H_{FS}$ )

$$H' = -\frac{e}{2m_e c} (\hat{L}_z + 2\hat{S}_z) B.$$

In the coupled basis  $|n, \ell, j, m_j\rangle$  (formula in §9.4): **Example values:**  $s_{1/2}$ :  $g_J = 2$ ;  $p_{1/2}$ :  $g_J = 2/3$ ;  $p_{3/2}$ :  $g_J = 4/3$ ;  $d_{3/2}$ :  $g_J = 4/5$ ;  $d_{5/2}$ :  $g_J = 6/5$ .

### 25.2 Strong-Field (Paschen-Back)

$H_Z \gg H_{FS}$ ; use uncoupled basis  $|n, \ell, m_\ell, m_s\rangle$ :

$$\Delta E = (m_\ell + 2m_s)\mu_B B + E_{FS}(m_\ell, m_s)$$

where the residual fine-structure correction ( $\ell \neq 0$ ) is:

$$E_{FS} = \frac{\alpha^4 m_e c^2}{2n^3} \left[ \frac{3}{4n} - \frac{\ell(\ell+1) - m_\ell m_s}{\ell(\ell+\frac{1}{2})(\ell+1)} \right]$$

The SO piece alone is  $\propto \frac{m_\ell m_s}{\ell(\ell+\frac{1}{2})(\ell+1)}$ ; consistent with  $\langle 1/r^3 \rangle = \frac{1}{n^3 a_0^3 \ell(\ell+\frac{1}{2})(\ell+1)}$ .

### 25.3 Intermediate Field

Diagonalise  $H_{FS} + H_Z$  in the coupled  $|j, m_j\rangle$  basis.  $H_Z \propto L_z + 2S_z$  connects  $j$  with  $j \pm 1$ .

**Example:**  $n = 2, \ell = 1$ . In the  $m_j = \frac{1}{2}$  subspace  $\left\{ \left| \frac{3}{2}, \frac{1}{2} \right\rangle, \left| \frac{1}{2}, \frac{1}{2} \right\rangle \right\}$ : Diagonal  $H_{FS}$ :  $E_{3/2}^{(0)}$  and  $E_{1/2}^{(0)}$ . Off-diagonal  $H_Z$ :  $\langle 3/2 | L_z + 2S_z | 1/2 \rangle \neq 0$ .

$$H = \begin{pmatrix} E_{3/2}^{(0)} + \frac{2}{3}\beta & \frac{\sqrt{2}}{3}\beta \\ \frac{\sqrt{2}}{3}\beta & E_{1/2}^{(0)} + \frac{1}{3}\beta \end{pmatrix}$$

where  $\beta = \mu_B B$ . Eigenvalues from characteristic equation.

## 26 Perturbation: Anharmonic Oscillator

### 26.1 Quartic Perturbation: $H' = \lambda x^4$

$$\hat{H}_0 = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 x^2; \quad H' = \lambda x^4.$$

Using  $x = \ell_0(\hat{a} + \hat{a}^\dagger)$  with  $\ell_0 = \sqrt{\hbar/(2m\omega)}$ :

$$x^4 = \ell_0^4 (\hat{a} + \hat{a}^\dagger)^4$$

**First-order correction:**

Keep only terms with equal numbers of  $\hat{a}$  and  $\hat{a}^\dagger$  (diagonal):

$$\langle n | x^4 | n \rangle = \ell_0^4 (6n^2 + 6n + 3)$$

$$E_n^{(1)} = \lambda \ell_0^4 (6n^2 + 6n + 3) = \frac{3\lambda \hbar^2}{4m^2 \omega^2} (2n^2 + 2n + 1)$$

**Ground state:**  $E_0^{(1)} = \frac{3\lambda \hbar^2}{4m^2 \omega^2}$ .

**Second-order correction (ground state):**

$$E_0^{(2)} = -\frac{21\lambda^2\hbar^3}{8m^4\omega^5}$$

Only  $|2\rangle$  and  $|4\rangle$  contribute via non-zero matrix elements: (Use normal-ordering trick from §4.3: expand  $(\hat{a} + \hat{a}^\dagger)^4$ , commute all  $\hat{a}$  right via  $[\hat{a}, \hat{a}^\dagger] = 1$ , keep only terms connecting  $|0\rangle$  to  $|2\rangle$  or  $|4\rangle$ .)

$$\langle 0 | x^4 | 2 \rangle = 3\sqrt{2}\ell_0^4, \quad \langle 0 | x^4 | 4 \rangle = \sqrt{24}\ell_0^4$$

**26.2 Cubic Perturbation:  $H' = \lambda x^3$** 

$$x^3 = \ell_0^3(\hat{a} + \hat{a}^\dagger)^3.$$

$$\langle n | x^3 | n \rangle = 0 \quad \forall n \quad \Rightarrow \quad E_n^{(1)} = 0$$

**Second-order (ground state):** Only  $\Delta n = \pm 1, \pm 3$  allowed by  $x^3$ .

$$E_0^{(2)} = \sum_k \frac{|\langle k | x^3 | 0 \rangle|^2}{E_0 - E_k} = \frac{|\langle 1 | x^3 | 0 \rangle|^2}{-\hbar\omega} + \frac{|\langle 3 | x^3 | 0 \rangle|^2}{-3\hbar\omega}$$

Substituting matrix elements ( $3\ell_0^3$  and  $\sqrt{6}\ell_0^3$ ):

$$E_0^{(2)} = \lambda^2 \ell_0^6 \left[ \frac{9}{-\hbar\omega} + \frac{6}{-3\hbar\omega} \right] = -\frac{11\lambda^2\hbar^2}{8m^3\omega^4}$$

$$E_0^{(2)} = -\frac{11\lambda^2\hbar}{8m^3\omega^4} \cdot \frac{\hbar}{1}$$

**General  $n$ :**

$$E_n^{(2)} = -\frac{\lambda^2 \ell_0^6}{\hbar\omega} [30n^2 + 30n + 11] \cdot \frac{1}{4}$$

**26.3 Linear Perturbation:  $H' = \lambda x$** 

$$E_n^{(1)} = \lambda \langle n | x | n \rangle = 0 \text{ (parity)}.$$

**Exact:** Completing the square in  $\hat{H}$ :

$$\hat{H} = \frac{p^2}{2m} + \frac{m\omega^2}{2} \left( x + \frac{\lambda}{m\omega^2} \right)^2 - \frac{\lambda^2}{2m\omega^2}$$

$$E_n = \hbar\omega(n + \frac{1}{2}) - \frac{\lambda^2}{2m\omega^2}$$

Perturbation theory:  $E_n^{(2)} = -\frac{\lambda^2}{2m\omega^2}$  (exact to all orders!).

**27 Perturbation: Perturbed Harmonic Oscillator****27.1 Shifted Frequency:  $H' = \frac{1}{2}m\epsilon\omega^2 x^2$** 

**Exact:** new frequency  $\omega' = \omega\sqrt{1 + \epsilon}$ .

$$E_n = \hbar\omega\sqrt{1 + \epsilon}(n + \frac{1}{2})$$

**Perturbative:**  $E_n^{(1)} = \frac{\epsilon}{2}m\omega^2 \langle n | x^2 | n \rangle = \frac{\epsilon\hbar\omega}{4}(2n + 1)$ .

Check: Taylor  $\omega\sqrt{1 + \epsilon} \approx \omega(1 + \frac{\epsilon}{2}) \Rightarrow$  agrees.

**27.2 Electric Field on Charged HO:  $H' = -qEx$** 

This is a constant force (linear perturbation). Exact solution via completing the square:

$$E_n = \hbar\omega(n + \frac{1}{2}) - \frac{q^2 E^2}{2m\omega^2}$$

$$E_n^{(1)} = 0; \quad E_n^{(2)} = -\frac{q^2 E^2}{2m\omega^2} \text{ (exact)}.$$

**27.3 Coupled Oscillators as Perturbation**

$$H_0 = \hbar\omega(n_1 + n_2 + 1); \quad H' = \lambda x_1 x_2.$$

Using  $x_i = \ell_0(\hat{a}_i + \hat{a}_i^\dagger)$ :

$$H' = \lambda \ell_0^2 (\hat{a}_1 + \hat{a}_1^\dagger)(\hat{a}_2 + \hat{a}_2^\dagger)$$

**First-order:**  $E_{n_1, n_2}^{(1)} = 0$  (all matrix elements off-diagonal).

**Second-order (ground state  $|0, 0\rangle$ ):**

$$E_{0,0}^{(2)} = \frac{|\langle 1, 1 | H' | 0, 0 \rangle|^2}{-2\hbar\omega} = -\frac{\lambda^2 \hbar}{8m^2\omega^3}$$

**Exact:** Normal mode frequencies  $\omega_\pm = \sqrt{\omega^2 \pm \lambda/m}$ .

**27.4 Momentum Perturbation:  $H' = \gamma p$** 

$$E_n^{(1)} = \gamma \langle n | p | n \rangle = 0.$$

Exact (boost frame):  $E_n = \hbar\omega(n + \frac{1}{2}) - \frac{\gamma^2 m}{2}$  (completing the square in  $p$ ).

**28 More Perturbation Problems****28.1 Infinite Square Well:  $H' = V_0$  (half-well)**

$H' = V_0$  for  $0 < x < L/2$ .

$$E_n^{(1)} = \frac{2}{L} \int_0^{L/2} V_0 \sin^2\left(\frac{n\pi x}{L}\right) dx$$

Evaluate using  $\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$ :

$$E_n^{(1)} = \frac{V_0}{2} - \frac{V_0}{\pi n} \sin\left(\frac{n\pi}{2}\right) \cdot \begin{cases} 0 & n \text{ even} \\ (-1)^{(n-1)/2} & n \text{ odd} \end{cases}$$

For even  $n$ :  $E_n^{(1)} = V_0/2$ . For  $n = 1$ :  $E_1^{(1)} = \frac{V_0}{2} - \frac{V_0}{\pi}$ .

**28.2 Infinite Square Well:  $H' = \alpha \delta(x - L/2)$** 

$$E_n^{(1)} = \alpha |\psi_n(L/2)|^2 = \frac{2\alpha}{L} \sin^2\left(\frac{n\pi}{2}\right) = \begin{cases} 2\alpha/L & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

Even- $n$  states have a node at  $L/2$  and are unaffected.

**28.3 2D HO:  $H' = \lambda xy$** 

Ground state ( $n_x = n_y = 0$ ) is non-degenerate:  $E_0^{(1)} = \langle 0, 0 | xy | 0, 0 \rangle = 0$  (integrand  $xe^{-x^2} \cdot ye^{-y^2}$  is odd).

First excited level ( $N = 1$ ): Degenerate states  $|1, 0\rangle, |0, 1\rangle$ . Diagonal terms zero (odd parity). Off-diagonal:

$$V_{12} = \lambda \langle 1, 0 | xy | 0, 1 \rangle = \lambda \langle 1 | x | 0 \rangle \langle 0 | y | 1 \rangle = \frac{\lambda\hbar}{2m\omega}$$

$$W = \frac{\lambda\hbar}{2m\omega} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Rightarrow E^{(1)} = \pm \frac{\lambda\hbar}{2m\omega}$$

Good states:  $|\pm\rangle = \frac{1}{\sqrt{2}}(|1, 0\rangle \pm |0, 1\rangle)$ .

**Exact:** Rotate  $45^\circ$ :  $x' = (x + y)/\sqrt{2}$ ,  $y' = (x - y)/\sqrt{2}$ :  $\omega_\pm = \sqrt{\omega^2 \pm \lambda/m}$ .

**28.4 Spin in Magnetic Field:  $H' = \alpha S_x$** 

$$H_0 = \omega_0 S_z; \quad H' = \alpha S_x = \frac{\alpha\hbar}{2} \sigma_x.$$

$$\text{Full Hamiltonian: } H = \frac{\hbar}{2} \begin{pmatrix} \omega_0 & \alpha \\ \alpha & -\omega_0 \end{pmatrix}.$$

**Exact eigenvalues:**  $E_\pm = \pm \frac{\hbar}{2} \sqrt{\omega_0^2 + \alpha^2}$ .

**Perturbative** ( $\alpha \ll \omega_0$ ):  $E_+^{(2)} = \frac{\alpha^2 \hbar}{4\omega_0}$ ;  $E_-^{(2)} = -\frac{\alpha^2 \hbar}{4\omega_0}$ .

**28.5 Hydrogen:  $H' = \alpha r^2$  (Isotropic Perturbation)**

$$E_n^{(1)} = \alpha \langle r^2 \rangle_{n\ell} = \alpha \frac{a_0^2 n^2}{2} [5n^2 + 1 - 3\ell(\ell + 1)]$$

Lifts the  $\ell$ -degeneracy of hydrogen (states with different  $\ell$  get different shifts).

**28.6 Gravitational Perturbation on HO:  $H' = mgx$** 

Same as linear perturbation ( $\lambda = mg$ ). Exact:

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right) - \frac{mg^2}{2\omega^2}$$

**Convention note:**  $H' = mgx$  suits a vertical HO where  $x$  is the vertical displacement. For a horizontal HO in gravity, the relevant axis may be  $y$  or  $z$ . The physics (shifted equilibrium, same energy gaps) is identical regardless of axis label.

## 29 Hooke's Atom (Two-Electron Harmonic Trap)

By defining the center-of-mass and relative coordinates as  $\mathbf{R} = \frac{1}{2}(\mathbf{r}_1 + \mathbf{r}_2)$  and  $\mathbf{u} = \mathbf{r}_2 - \mathbf{r}_1$ , we invoke  $\nabla_1^2 + \nabla_2^2 = \frac{1}{2}\nabla_R^2 + 2\nabla_u^2$  and  $r_1^2 + r_2^2 = 2R^2 + \frac{1}{2}u^2$ . The Hamiltonian separates exactly:  $\hat{H} = \hat{H}_{CM} + \hat{H}_{rel}$ .

$$\hat{H} = \sum_{i=1}^2 \left( -\frac{1}{2}\nabla_i^2 + \frac{1}{2}kr_i^2 \right) + \frac{1}{|\mathbf{r}_1 - \mathbf{r}_2|}$$

$$\hat{H}_{CM} = -\frac{1}{4}\nabla_R^2 + kR^2 \implies E_R = \frac{3}{2}\sqrt{k}, \quad \chi(\mathbf{R}) \propto e^{-\sqrt{k}R^2}$$

$$\hat{H}_{rel} = -\nabla_u^2 + \frac{k}{4}u^2 + \frac{1}{u} \implies E_\ell = \frac{2\ell + 5}{4(\ell + 1)} \quad \text{for } k = \frac{1}{2(\ell + 1)}$$

**Exact Ground State Solution** ( $k = 1/4, \ell = 0$ ):

Analytic solutions for  $\hat{H}_{rel}$  exist for discrete values of  $k$  where the Frobenius series terminates. For  $k = 1/4$ , the relative energy is  $E_u = 5/4$ , yielding a total ground-state energy  $E_{tot} = E_R + E_u = 3/4 + 5/4 = 2$ . The exact, fully normalized two-electron wavefunction is:

$$\Psi(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{28\pi^{5/2} + 5\pi^3}} \left( 1 + \frac{|\mathbf{r}_1 - \mathbf{r}_2|}{2} \right) e^{-\frac{1}{4}(r_1^2 + r_2^2)}$$

### 29.1 Quadratic Perturbation on Hydrogen: $V = \lambda z^2$

**Ground State** ( $n = 1$ ): Isotropic symmetry ( $l = 0$ ) implies  $\langle z^2 \rangle = \frac{1}{3}\langle r^2 \rangle$ . With  $\langle r^2 \rangle_{100} = 3a_0^2$ :

$$E_1^{(1)} = \langle 100 | \lambda z^2 | 100 \rangle = \frac{\lambda}{3} \langle 100 | r^2 | 100 \rangle = \lambda a_0^2$$

**Excited States** ( $n = 2$ ): The operator  $z^2$  has **even parity**, meaning  $\langle 200 | z^2 | 21m \rangle = 0$  (no  $2s$ - $2p$  mixing). Since  $\Delta m = 0$ , the  $4 \times 4$  perturbation matrix is completely diagonal. Using  $\langle r^2 \rangle_{nl} = \frac{a_0^2 n^2}{2} [5n^2 + 1 - 3l(l+1)]$ , we have  $\langle r^2 \rangle_{20} = 42a_0^2$  and  $\langle r^2 \rangle_{21} = 30a_0^2$ .

$2s$  ( $l = 0, m = 0$ ): Spherically symmetric.  $E_{200}^{(1)} = \frac{\lambda}{3} \langle r^2 \rangle_{20} = 14\lambda a_0^2$

$2p_z$  ( $l = 1, m = 0$ ): Aligned with  $z$ -axis.  $E_{210}^{(1)} = \lambda \langle r^2 \rangle_{21} \int |Y_{10}|^2 \cos^2 \theta d\Omega = 30\lambda a_0^2 \left( \frac{3}{5} \right) = 18\lambda a_0^2$

$2p_{x,y}$  ( $l = 1, m = \pm 1$ ): Equatorial concentration.  $E_{21\pm 1}^{(1)} = \lambda \langle r^2 \rangle_{21} \int |Y_{1\pm 1}|^2 \cos^2 \theta d\Omega = 30\lambda a_0^2 \left( \frac{1}{5} \right) = 6\lambda a_0^2$

**Splitting Summary:**

$$E_{210}^{(1)} = 18\lambda a_0^2 > E_{200}^{(1)} = 14\lambda a_0^2 > E_{21\pm 1}^{(1)} = 6\lambda a_0^2$$

## 30 Summary of Perturbation Matrix Elements (HO)

Key matrix elements for harmonic oscillator perturbations ( $\ell_0 = \sqrt{\hbar/2m\omega}$ ):

$$\langle m | x | n \rangle: \ell_0 (\sqrt{n} \delta_{m,n-1} + \sqrt{n+1} \delta_{m,n+1}).$$

$$\langle m | x^2 | n \rangle: \ell_0^2 (\sqrt{n(n-1)} \delta_{m,n-2} + (2n+1) \delta_{m,n} + \sqrt{(n+1)(n+2)} \delta_{m,n+2}).$$

$$\langle m | x^3 | n \rangle:$$

$$\ell_0^3 [\sqrt{n(n-1)(n-2)} \delta_{m,n-3} + 3n^{3/2} \delta_{m,n-1} + 3(n+1)^{3/2} \delta_{m,n+1} + \sqrt{(n+1)(n+2)(n+3)} \delta_{m,n+3}]$$

$$\langle n | x^4 | n \rangle: \ell_0^4 (6n^2 + 6n + 3).$$

$$\langle n | p^2 | n \rangle: \frac{m\hbar\omega}{2} (2n+1).$$

$$\langle n | p^4 | n \rangle: \frac{3m^2 \hbar^2 \omega^2}{4} (2n^2 + 2n + 1).$$

## 31 Exam Archetype Problems (by Topic)

### 31.1 A. Symmetry & Degeneracy Proofs

#### A1. Spin-Rotation Symmetry $\Rightarrow$ Degeneracy

**Problem.**  $S = \frac{1}{2}$ ,  $[H, S_a] = 0 \forall a$ . Show all levels  $\geq 2$ -fold degenerate.

**Solution.** Let  $|v, \downarrow\rangle$  be eigenstate with energy  $E$ . Since  $[H, S_+ ] = 0$ :  $H(S_+ |v, \downarrow\rangle) = E(S_+ |v, \downarrow\rangle)$ . For  $S = \frac{1}{2}$ :  $S_+ |v, \downarrow\rangle \propto |v, \uparrow\rangle \neq 0$  and  $\perp |v, \downarrow\rangle$ .  $\implies$  Every level has  $\geq 2$  states.  $\checkmark$

#### A2. Kramers Degeneracy ( $[H, T] = 0$ , half-integer spin)

**Problem.**  $S = \frac{1}{2}$ ,  $[H, T] = 0$ . Show all levels  $\geq 2$ -fold degenerate.

**Solution.**  $T|v\rangle$  is eigenstate with same  $E$ . Assume  $T|v\rangle = c|v\rangle$ . Apply  $T$  again:  $T^2|v\rangle = |c|^2|v\rangle$ . But  $T^2 = -1$  for half-integer spin  $\implies |c|^2 = -1$ . **Contradiction.**  $\implies |v\rangle$  and  $T|v\rangle$  are linearly independent (in fact orthogonal).  $\checkmark$

**Key:**  $T^2 = (-1)^{2S}$ . Integer spin:  $T^2 = +1$ , no forced degeneracy. Half-integer:  $T^2 = -1$ , Kramers pairs.

#### A3. Non-degenerate + $[H, P] = 0 \Rightarrow$ Parity Eigenstate

**Solution.**  $P|\Psi_n\rangle$  has energy  $E_n$ . Non-degenerate  $\implies P|\Psi_n\rangle = \lambda|\Psi_n\rangle$ .  $P^2 = \mathbb{I} \implies \lambda^2 = 1 \implies \lambda = \pm 1$ .  $\checkmark$

#### A4. Non-degenerate + $[H, T] = 0 \Rightarrow$ Real Wavefunction

**Solution.**  $T|\Psi_n\rangle = c|\Psi_n\rangle$ ,  $|c| = 1$ . In position space ( $T\psi = \psi^*$ ):  $\psi^*(x) = c\psi(x)$ . Write  $\psi = |\psi|e^{i\theta}$ :  $e^{-2i\theta(x)} = c \implies \theta = \text{const.}$  Define  $\tilde{\psi} = e^{-i\theta}\psi \implies \tilde{\psi} \in \mathbb{R}$ .  $\checkmark$

#### A5. Time Reversal of Quadratic Spin Hamiltonian

$H = \alpha J_z^2 + \beta(J_x^2 - J_y^2)$ . Is  $[H, T] = 0$ ?  $T\vec{J}T^{-1} = -\vec{J} \implies TJ_i^2T^{-1} = J_i^2$ . All terms even  $\implies$  **Yes**. Integer  $J$ : eigenstates map to themselves (up to phase); no forced degeneracy.

### 31.2 B. Angular Momentum & $J = 1$ Matrix Problems

#### B1. $\langle J_x \rangle$ and $\langle J_x^2 \rangle$ in $|j, m\rangle$

$J_x = \frac{1}{2}(J_+ + J_-)$ . Since  $J_\pm |j, m\rangle \propto |j, m \pm 1\rangle \perp |j, m\rangle$ :  $\langle J_x \rangle = 0$ .

$J_x^2 = \frac{1}{4}(J_+^2 + J_-^2 + J_+J_- + J_-J_+)$ .  $J_\pm^2$  terms  $\rightarrow 0$  (shift  $m$  by 2). Use

$$J_+J_- + J_-J_+ = 2(J^2 - J_z^2): \quad \langle J_x^2 \rangle = \frac{\hbar^2}{2} [j(j+1) - m^2].$$

#### B2. Diagonalise $H = \alpha J_z^2 + \beta(J_x^2 - J_y^2)$ for $J = 1$

**Key identity:**  $J_x^2 - J_y^2 = \frac{1}{2}(J_+^2 + J_-^2)$  (shifts  $m$  by  $\pm 2$ ).

Basis  $\{|1\rangle, |0\rangle, |-1\rangle\}$ .  $J_z^2 = \hbar^2 \text{diag}(1, 0, 1)$ .  $J_+^2 |-1\rangle = 2\hbar^2 |1\rangle$ ,  $J_-^2 |1\rangle = 2\hbar^2 |-1\rangle$ .

$$\text{Matrix (units } \hbar = 1): H = \begin{pmatrix} \alpha & 0 & \beta \\ 0 & 0 & 0 \\ \beta & 0 & \alpha \end{pmatrix}.$$

**Eigenvalues:**  $E = 0$  ( $|0\rangle$ ),  $\alpha + \beta$  ( $\frac{1}{\sqrt{2}}(|1\rangle + |-1\rangle)$ ),  $\alpha - \beta$  ( $\frac{1}{\sqrt{2}}(|1\rangle - |-1\rangle)$ ).

**Template:** Any  $H$  quadratic in  $J_i$  for  $J = 1$  reduces to at most  $3 \times 3$  matrix;  $|0\rangle$  often decouples.

#### B3. Particle Decay & Angular Momentum Addition

**Problem.**  $A \rightarrow B + C$  with  $A$  at rest. (a) conservation law; (b) allowed  $L$ ; (c) can  $n \rightarrow p + e^-$ ? (d) Given  $S_B = \frac{1}{2}, S_C = 0$ ,  $A$  maximally polarised ( $m_A = S_A$ ); if  $P(m_B = \frac{1}{2}) = 1/5$ ,  $P(m_B = -1/2) = 4/5$ , find  $L, S_A$ .

(a)  $\vec{S}_A = \vec{L} + \vec{S}_B + \vec{S}_C$  (total AM). (b)  $S_{BC} = S_B + S_C$ , so  $|S_A - S_{BC}| \leq L \leq S_A + S_{BC}$ . (c) No:  $S_n = 1/2$ ;  $S_p = S_e = 1/2 \implies S_{BC} \in \{0, 1\}$ ;  $L \in \mathbb{Z}$  so final  $J = L + S_{BC}$  integer, cannot equal half-integer  $S_A$ . (d) Here  $S_{BC} = 1/2$ , so  $J = S_A = L \pm 1/2$ . Max polarisation implies  $S_A = L - 1/2$  (otherwise  $P(m_B = 1/5)$  impossible). Write

$$|S_A, S_A\rangle = \alpha |L-1, 1/2\rangle + \beta |L, -1/2\rangle, \quad \beta = -\sqrt{2L}\alpha, \alpha^2(1+2L) = 1.$$

Then  $P(m_B = 1/2) = \alpha^2 = 1/(2L+1) = 1/5 \implies L = 2$ , hence  $S_A = 3/2$ . If given spin measurement probabilities in this coupled state, use  $2L = P_1/P_0$  to find  $L$  directly.

### 31.3 C. Hydrogen-like Atoms & Shell Filling

#### C1. Spin-3/2 Particle ("Omicron") with Spin-Orbit

$$H = \frac{p^2}{2m} - \frac{Ze^2}{r} + \frac{\alpha R_y}{\hbar^2} \vec{L} \cdot \vec{S}, \quad s = 3/2.$$

**Recipe:**  $E_n^{(0)} = -Z^2/n^2 \text{ Ry}$ .  $\Delta E_{LS} = \frac{\alpha}{2} [j(j+1) - l(l+1) - \frac{15}{4}] \text{ Ry}$ .

For  $l = 0$ :  $j = 3/2$  only,  $\Delta E = 0$ . Degen = 4.

For  $l = 1$ :  $j \in \{\frac{1}{2}, \frac{3}{2}, \frac{5}{2}\}$ .  $j = \frac{1}{2}$ :  $\lambda = -2.5$ , deg 2.  $j = \frac{3}{2}$ :  $\lambda = -1$ , deg 4.  $j = \frac{5}{2}$ :  $\lambda = +1.5$ , deg 6.

**Shell filling (Aufbau):** Each  $j$ -subshell holds  $2j+1$  particles. Order by energy (lowest first). Example for  $Z = 10$ , 10 particles:  $1s_{3/2}(4) + 2s_{3/2}(4) + 2p_{1/2}(2) = 10$ .  $E_{tot} = 4(-Z^2) + 4(-Z^2/4) + 2(-Z^2/4 - 2.5\alpha) = -550 - 5\alpha \text{ Ry}$ .



### 31.4 D. Identical Particles

#### D1. Two Fermions in Same Shell ( $j_1 = j_2 = j$ )

Spatial symmetric (same orbital)  $\implies$  spin must be **antisymmetric**.  $j \otimes j$ : CG symmetry  $\langle jj m_1 m_2 | JM \rangle = (-1)^{2j-J} \langle jj m_2 m_1 | JM \rangle$ . Antisymmetric  $\iff 2j - J$  odd  $\iff J$  even (for half-integer  $j$ ).

**Example:**  $j = 3/2$ : allowed  $J \in \{0, 2\}$  (antisym).  $|J = 2, M = 2\rangle = \frac{1}{\sqrt{2}} \left( \left| \frac{3}{2}, \frac{1}{2} \right\rangle - \left| \frac{1}{2}, \frac{3}{2} \right\rangle \right)$ .

#### D2. Delta Baryons ( $uuu$ , $S = 3/2$ )

3 quarks, color singlet (antisym). Spatial  $L = 0$  (sym).  $\implies$  Spin $\times$ Isospin must be symmetric.  $\Delta$ :  $I = 3/2$  (sym)  $\times$   $S = 3/2$  (sym) = sym.  $\checkmark$

Wavefunctions ( $S_z = 3/2$ , all spins up):  $\Delta^{++} = |uuu\rangle |\uparrow\uparrow\uparrow\rangle$ ,  $\Delta^+ = \frac{1}{\sqrt{3}}(|uud\rangle + |udu\rangle + |duu\rangle) |\uparrow\uparrow\uparrow\rangle$ .

**Magnetic moments:**  $\mu = \sum_i \mu_i$ .  $\mu_u = -2\mu_d$ .  $\Delta^{++} : 3\mu_u$ ,  $\Delta^+ : \frac{3}{2}\mu_u$ ,  $\Delta^0 : 0$ ,  $\Delta^- : -\frac{3}{2}\mu_u$ . Ratios:  $1 : \frac{1}{2} : 0 : -\frac{1}{2}$ .

**Can  $\Delta^{++}$  have  $S = 1/2$ ?** No.  $uuu$  flavor is totally symmetric.  $L = 0$  symmetric. Need spin symmetric. For 3 spins- $\frac{1}{2}$ :  $\frac{1}{2} \otimes \frac{1}{2} \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2} \oplus \frac{1}{2}$ . Only  $S = 3/2$  is totally symmetric;  $S = 1/2$  states have mixed symmetry. **Impossible.**

#### D3. Scattering of Identical Bosons ( $S = 1$ )

$1 \otimes 1 = 2_S \oplus 1_A \oplus 0_S$ . Total wf must be symmetric (bosons).

$S = 0, 2$  (sym spin)  $\implies$  sym spatial:  $\psi_S \sim [f(\theta) + f(\pi - \theta)] \frac{e^{ikr}}{r}$ .

$S = 1$  (antisym spin)  $\implies$  antisym spatial:  $\psi_A \sim [f(\theta) - f(\pi - \theta)] \frac{e^{ikr}}{r}$ .

$\frac{d\sigma}{d\Omega} \Big|_{S=0,2} = |f(\theta) + f(\pi - \theta)|^2$ ,  $\frac{d\sigma}{d\Omega} \Big|_{S=1} = |f(\theta) - f(\pi - \theta)|^2$ .

At  $\theta = 90^\circ$ :  $f(\pi/2) = f(\pi/2) \implies$  antisym  $\rightarrow 0$ .  $S = 1$  forbidden at  $90^\circ$ .

**Unpolarized:**  $\frac{d\sigma}{d\Omega} = \frac{1}{9}(6\sigma_{\text{sym}} + 3\sigma_{\text{anti}}) = \frac{2}{3}|f + f'|^2 + \frac{1}{3}|f - f'|^2$  where  $f' = f(\pi - \theta)$ .

### 31.5 Perturbation Theory Archetypes

#### E1. Contact Interaction $W = w_0[\delta^{(3)}p_z + p_z\delta^{(3)}] \frac{\hbar}{2}$

**Selection rules:**  $\delta(r)$  needs  $\psi(0) \neq 0$  ( $l = 0$ ).  $p_z \sim Y_{1,0}$  needs  $l = 1$ ,  $\Delta m = 0$ .

Combined:  $[l + l' = 1, m = m' = 0]$  (one state  $s$ , other  $p$ ).

$n = 1$ :  $E^{(1)} = \langle 100 | W | 100 \rangle = 0$  (need  $l + l' = 1$ , but  $0 + 0 = 0$ ).

$n = 2$ : Degenerate PT in  $\{|200\rangle, |210\rangle, |211\rangle, |21-1\rangle\}$ . Only  $|200\rangle \leftrightarrow |210\rangle$

couples (both others have  $m \neq 0$ ).  $W = \begin{pmatrix} 0 & V & 0 & 0 \\ V^* & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$  with  $V = -i \frac{\hbar^2 w_0}{32\pi a_0^4}$ .

$E^{(1)} = \pm|V| = \pm \frac{\hbar^2 w_0}{32\pi a_0^4}$ . Good states:  $\frac{1}{\sqrt{2}}(|2s\rangle \mp i|2p_0\rangle)$ .  $|21 \pm 1\rangle$ : correction = 0.

#### E2. Quantum Top in $\vec{B}$ (2nd-order)

$H_0 = \frac{L^2}{2I} - \gamma B_0 L_z$ .  $E_{lm} = \frac{\hbar^2 l(l+1)}{2I} - \gamma B_0 \hbar m$ .

Perturbation  $V = -\gamma B_1 L_x = -\frac{\gamma B_1}{2}(L_+ + L_-)$ .  $E^{(1)} = \langle m | L_x | m \rangle = 0$ .

$E^{(2)}$ :  $L_x$  connects  $m \rightarrow m \pm 1$ . Energy gaps =  $\mp \gamma B_0 \hbar$ .  $|V_{m,m+1}|^2 = \frac{\gamma^2 B_1^2 \hbar^2}{4} [l(l+1) - m(m+1)]$ .

$\Delta E = \frac{\gamma^2 B_1^2 \hbar}{4\gamma B_0 \hbar} [C_+^2 - C_-^2]$  where  $C_+^2 - C_-^2 = -2m$ .  $\Delta E^{(2)} = -\frac{\gamma B_1^2 \hbar m}{2B_0}$ .

**Check:** Exact  $E = -\gamma \sqrt{B_0^2 + B_1^2} \hbar m \approx -\gamma B_0 \hbar m (1 + \frac{B_1^2}{2B_0^2})$ . Matches.

#### E3. Spin-1 in Well with Perturbation

$V = \hbar(r)\hat{n} \cdot \vec{S}$ ,  $r \leq R$ ,  $\infty$  outside. Choose  $\hat{n} = \hat{z}$ .

**Symmetries:**  $E \checkmark, \vec{p} \times, \vec{L} \checkmark, \vec{J} \times, P \checkmark, T \times$  ( $\vec{S} \rightarrow -\vec{S}$ ).

**Spectrum** ( $\hbar = 1$ ):  $H = \frac{p^2}{2m} + S_z + V_{\text{well}}$ . Separates:  $E_{n,\mu} = \frac{\hbar^2 \pi^2 n^2}{2mR^2} + \mu \hbar$ ,  $\mu \in \{-1, 0, 1\}$ .

**Perturbation**  $\hbar = 1 + \lambda \cos(\pi r/R)$ :  $E^{(1)} \propto \int_0^R \sin^2(\pi r/R) \cos(\pi r/R) dr = 0$  (odd power integral).  $E^{(2)}$ : Only  $n = 2$  contributes (use  $\sin x \cos x = \frac{1}{2} \sin 2x$ ).

$\Delta E = -\frac{\mu^2 \lambda^2}{12E_0}$  where  $E_0 = \frac{\hbar^2 \pi^2}{2mR^2}$ .

### 31.6 F. Time-Dependent Perturbation Theory

#### F1. Exponentially Decaying Field on H Atom

$\vec{E} = \frac{\phi_0}{a_0} \hat{z} e^{-t/\tau}$  for  $t \geq 0$ . Initial state  $|1, 0, 0\rangle$ .

**Selection rules:**  $V \propto z = r \cos \theta \propto Y_{1,0}$ .  $\Delta l = \pm 1$ ,  $\Delta m = 0$ . From  $l = 0, m = 0$ : allowed  $\rightarrow |n, 1, 0\rangle$ .

**Amplitude** ( $|i\rangle \rightarrow |f\rangle = |2, 1, 0\rangle$ ):  $c_f = \frac{-i\mathcal{M}}{\hbar} \int_0^t e^{(i\omega_{fi} - 1/\tau)t'} dt'$ .

As  $t \rightarrow \infty$ :  $c_f = \frac{\mathcal{M}}{\hbar} \frac{1}{\omega_{fi} + i/\tau}$ .  $P_{210} = \frac{|\mathcal{M}|^2}{\hbar^2(\omega_{fi}^2 + 1/\tau^2)}$ ,  $P_{21\pm 1} = 0$ .

**Limits:**  $\tau \rightarrow 0$  (sudden):  $P \propto \tau^2 \rightarrow 0$  (pulse too fast).  $\tau \rightarrow \infty$ :  $P \rightarrow |\mathcal{M}|^2/(\hbar\omega_{fi})^2$  (quasi-static mixing  $\sim |V/\Delta E|^2$ ).

**If initial  $|2, 0, 0\rangle$ :** Target  $|2, 1, 0\rangle$  is degenerate ( $\omega_{fi} = 0$ ). Denominator  $\rightarrow 1/\tau$  only.  $P \propto \tau^2 \rightarrow \infty$ . Must use **degenerate PT** (linear Stark effect). Stark eigenstates:  $|\pm\rangle = \frac{1}{\sqrt{2}}(|200\rangle \mp |210\rangle)$ .

### 31.7 G. 3D Potentials & Bound States

#### G1. 3D Finite Spherical Well ( $\ell = 0$ )

**Problem.**  $V(r) = -V_0$  for  $r \leq a$ , else 0. Find ground state and min  $V_0$ .

**Solve.** With  $u = r\psi$ :  $-\frac{\hbar^2}{2m}u'' + V(r)u = Eu$ ,  $u(0) = u(\infty) = 0$ . For  $r > a$ :  $u'' = -\frac{2mE}{\hbar^2}u$ ;  $E > 0$ :  $u = A \sin kr + B \cos kr$ ,  $E < 0$ :  $u = Ce^{-kr} + De^{kr}$ .

Ground state ( $E < 0$ ):  $u_{in} = A \sin qr$  with  $q = \frac{\sqrt{2m(V_0 - |E|)}}{\hbar}$ ;  $u_{out} = Ce^{-kr}$  ( $D = 0$ ).

$$\psi = \begin{cases} A \sin(qr)/r, & r \leq a \\ Ce^{-kr}/r, & r > a \end{cases}$$

Match at  $r = a$ :  $\frac{u'}{u} = q \cot qa = -\kappa$ .

Threshold  $E \rightarrow 0^-$  ( $\kappa \rightarrow 0$ ) gives  $qa = \pi/2$ , hence

$$V_0^{\min} = \frac{\pi^2 \hbar^2}{8ma^2}$$

(A 3D well needs a minimum depth/size to bind, unlike 1D.)

### 31.8 G. Sudden & Adiabatic Approximation

The Sudden Approximation applies when the perturbation occurs much faster than the system's characteristic response time (wavefunction period).

$$\tau \ll T_{\text{char}} \approx \frac{1}{|\omega_{ni}|} = \frac{\hbar}{|E_n - E_i|}$$

#### G1. Spherical Well $R \rightarrow 2R$

**Energies** ( $L = 0$ ):  $u(r) = rR(r)$ ,  $u'' = -k^2u$ , BCs  $u(0) = u(R) = 0$ .

$$E_n = \frac{\hbar^2 \pi^2 n^2}{2mR^2}, \quad u_n = \sqrt{\frac{2}{R}} \sin \frac{n\pi r}{R}$$

**Sudden** ( $R \rightarrow 2R$ ):  $P = |\langle \psi_1^{(2R)} | \phi_1^{(R)} \rangle|^2$ .  $\langle \cdot | \cdot \rangle = \frac{\sqrt{2}}{R} \int_0^R \sin\left(\frac{\pi r}{2R}\right) \sin\left(\frac{\pi r}{R}\right) dr$ .

Substitution  $x = \pi r/2R$ , use  $\sin(2x) = 2 \sin x \cos x$ :  $= \frac{2\sqrt{2}}{\pi}$ .

$$2 \int_0^{\pi/2} \sin^2 x \cos x dx = \frac{4\sqrt{2}}{\pi} \cdot \frac{1}{3} = \frac{4\sqrt{2}}{3\pi}. \quad P = \frac{32}{9\pi^2} \approx 0.36$$

**Adiabatic:** System stays in  $n = 1$ .  $P = 1$ .

**Criterion:**  $\tau_{\text{exp}} \gg \hbar/\Delta E$  (adiabatic);  $\tau_{\text{exp}} \ll \hbar/\Delta E$  (sudden).  $\Delta E = E_2 - E_1 = \frac{3\hbar^2 \pi^2}{2mR^2}$ .

#### G2. Tritium $\beta$ -Decay ( $Z=1 \rightarrow Z=2$ )

Sudden change of nuclear charge. Electron stays in  $\psi_{1s}^{Z=1}$ . Project onto  $\psi_{nlm}^{Z=2}$  basis: **General overlap formula** (both states are 1s with different  $Z$ ):

$$\langle 1s, Z' | 1s, Z \rangle = \frac{8(ZZ')^{3/2}}{(Z + Z')^3}$$

**Derivation:**  $\langle 1s, Z' | 1s, Z \rangle = \int \psi_{1s}^{Z'} \psi_{1s}^Z d^3r$ ; both are  $\propto e^{-Zr/a_0}$ ; use  $\int_0^\infty r^2 e^{-ar} dr = 2/a^3$ . For  $Z = 1, Z' = 2$ : overlap =  $16\sqrt{2}/27$ ,  $P = \frac{512}{729} \approx 0.70$ . Selection:  $\Delta l = 0$  (scalar perturbation  $\implies$  only  $s$ -states populated).

### 31.9 H. Scattering Archetypes

#### H1. Separable Potential $V = \lambda |\chi\rangle \langle \chi|$

$\langle r | \chi \rangle \propto e^{-r^2/2R^2}$  (Gaussian).

**Born:**  $T_{fi} = \lambda \langle \mathbf{k}' | \chi \rangle \langle \chi | \mathbf{k} \rangle$ . FT of Gaussian = Gaussian:  $\langle \mathbf{k} | \chi \rangle \propto e^{-k^2 R^2/2}$ .  $\implies T_{fi} \propto \lambda e^{-(k^2 + k'^2)R^2/2}$ . Elastic ( $k = k'$ ):  $f \propto \lambda e^{-k^2 R^2}$ . **Isotropic** ( $\theta$ -independent).



$$\sigma = 4\pi|f|^2 \propto \lambda^2 e^{-2k^2 R^2}.$$

## H2. Two-Center Scattering

Target at  $\mathbf{0}$  and  $\mathbf{a}$ :  $V_{\text{tot}} = V(\mathbf{r}) + V(\mathbf{r}-\mathbf{a})$ . Shift theorem:  $f_{\text{tot}} = f_1(\theta)[1 + e^{iq \cdot \mathbf{a}}]$ . Interference zeros:  $\mathbf{q} \cdot \mathbf{a} = (2n+1)\pi$ . For  $\mathbf{a} = a\hat{z}$ :  $ka(1 - \cos \theta) = (2n+1)\pi$ . Form-factor zeros:  $\tan(qR) = qR$ .

### Born Approx & Structure Factor (Yukawa)

**Problem:**  $V(r) = \beta e^{-\mu r}/r$ , two centers at  $\mathbf{0}$  and  $\mathbf{x}$ , and a molecule with  $N$  particles distributed via Gaussian  $n(\mathbf{r}) \propto e^{-r^2/(2a^2)}$ .

$$f_0(\mathbf{q}) = -\frac{2m}{\hbar^2 q} \int_0^\infty r V(r) \sin(qr) dr. \text{ For Yukawa: } \int_0^\infty e^{-\mu r} \sin(qr) dr =$$

$$\frac{q}{\mu^2 + q^2} \implies f_0(\mathbf{q}) = -\frac{2m\beta}{\hbar^2(\mu^2 + q^2)}. d\sigma_0/d\Omega = |f_0|^2 = \frac{4m^2\beta^2}{\hbar^4(\mu^2 + q^2)^2}.$$

**Two-Center Scattering:**  $V_{\text{tot}} = V(\mathbf{r}) + V(\mathbf{r}-\mathbf{x})$ . Shift theorem:  $f_{\text{tot}}(\mathbf{q}) = f_0(\mathbf{q})S(\mathbf{q})$  where  $S(\mathbf{q}) = 1 + e^{-iq \cdot \mathbf{x}}$ . expanding the squared structure factor:  $|S(\mathbf{q})|^2 = |1 + e^{-iq \cdot \mathbf{x}}|^2 = (1 + e^{-iq \cdot \mathbf{x}})(1 + e^{iq \cdot \mathbf{x}}) = 4 \cos^2\left(\frac{\mathbf{q} \cdot \mathbf{x}}{2}\right)$ .  $\implies d\sigma/d\Omega = 4 \frac{d\sigma_0}{d\Omega} \cos^2\left(\frac{\mathbf{q} \cdot \mathbf{x}}{2}\right)$ .

**Continuous density** The molecular potential is the convolution of single-particle potential with density:  $V_{\text{mol}}(\mathbf{r}) = N(n * V)(\mathbf{r})$ .

By the convolution theorem the Born amplitude factorizes as  $f_{\text{mol}}(\mathbf{q}) = N f_0(\mathbf{q}) F_n(\mathbf{q})$ , with form factor  $F_n(\mathbf{q}) = \int n(\mathbf{r}) e^{-iq \cdot \mathbf{r}} d^3r$ .

For Gaussian  $n(\mathbf{r}) = (2\pi a^2)^{-3/2} \exp(-r^2/2a^2)$  one has  $F_n(\mathbf{q}) = \exp(-q^2 a^2/2)$ .

$$\text{Thus } \frac{d\sigma_{\text{mol}}}{d\Omega} = N^2 |f_0(\mathbf{q})|^2 \exp(-q^2 a^2) = N^2 \frac{d\sigma_0}{d\Omega} e^{-q^2 a^2}.$$

### 31.10 IV. Particle on Ring (R) in Field $\mathbf{B} = \beta \hat{z}$ (Symmetric Gauge)

Gauge  $\mathbf{A} = -\frac{1}{2}\beta(y\hat{x} - x\hat{y}) \implies \mathbf{B} = \beta\hat{z}$ . Ham:  $H = \frac{1}{2mR^2} \left(L_z - \frac{1}{2}q\beta R^2\right)^2$ .

**States:**  $\psi_n(\phi) = \frac{e^{in\phi}}{\sqrt{2\pi R}}$  ( $n \in \mathbb{Z}$ ). **Spectrum:**  $E_n(\beta) = \frac{1}{2mR^2} (\hbar n - \frac{1}{2}q\beta R^2)^2$ .

Coincides with free spectrum if  $\beta = \frac{2\hbar k}{qR^2}$  ( $k \in \mathbb{Z}$ ). **Dynamics:** Since  $[H(t), L_z] = 0$ , transitions are forbidden ( $P_{i \rightarrow f} = 0$ ). State  $n$  is conserved.

**Work/Energy:** For G.S. ( $n = 0$ ) ramped  $0 \rightarrow \beta_T$ :  $\Delta E = \frac{q^2 \beta_T^2 R^2}{8m}$ . **Work derivation:**  $\mathbf{E} = -\dot{\mathbf{A}}$ ;  $\langle \mathbf{v} \rangle_0 \approx -q\mathbf{A}/m$ .  $W = \int q\mathbf{E} \cdot \langle \mathbf{v} \rangle dt = \frac{q^2}{2m} \Delta(A^2) = \frac{q^2}{2m} (\frac{\beta_T R}{2})^2 = \Delta E$ .

### 31.11 J. Two-Spin Time Evolution

$$H(t) = g(t) \vec{S}_1 \cdot \vec{S}_2. \quad \vec{S}_1 \cdot \vec{S}_2 = \frac{1}{2}(S^2 - S_1^2 - S_2^2).$$

Coupled basis  $|S, M\rangle$  diagonalises  $H$ : Triplet  $S = 1$ :  $\langle \vec{S}_1 \cdot \vec{S}_2 \rangle = +\frac{\hbar^2}{4}$ .

$$\text{Singlet } S = 0: \langle \vec{S}_1 \cdot \vec{S}_2 \rangle = -\frac{3\hbar^2}{4}.$$

Initial  $|\uparrow\downarrow\rangle = \frac{1}{\sqrt{2}}(|T_0\rangle + |S_0\rangle)$ . Evolve: each component picks up phase  $e^{-\frac{i}{\hbar} \int_0^t E_S(t') dt'}$ . Phase difference  $\Delta\phi = \frac{1}{\hbar} \int g(t') dt' \cdot \hbar^2$ . Transition probability  $P_{|\uparrow\downarrow\rangle \rightarrow |\uparrow\uparrow\rangle} = \sin^2(\Delta\phi/2)$ .

### 31.12 K. Linear Stark Effect on Hydrogen ( $n = 2$ )

$H' = -eE_0 z$ . **Selection rules** (Wigner-Eckart):  $z \propto Y_{1,0}$  (rank-1 spherical tensor,  $q = 0$ , odd parity).  $\implies \Delta m = 0$ ,  $\Delta l = \pm 1$  (parity:  $l' \neq l$ ; triangle:  $|l - 1| \leq l' \leq l + 1$ ).

From  $|100\rangle$ : first-order vanishes ( $l' = l \pm 1$  forces  $l' = 1$ , no  $n = 1, l = 1$  state to mix).  $E_1^{(1)} = 0$ . Second-order:  $E_1^{(2)} < 0$  (polarisability).

**Degenerate subspace  $n = 2$ :**  $|211\rangle, |21-1\rangle$ : no coupling (need  $\Delta m = 0$ , no  $l = 0$  partner).  $E^{(1)} = 0$ .

$$|200\rangle \leftrightarrow |210\rangle \text{ couple. } \langle 200|z|210\rangle = \langle 200|r \cos \theta|210\rangle = -3a_0.$$

$$\text{Derivation: } z = r \sqrt{\frac{4\pi}{3}} Y_{1,0}. \text{ Radial: } \int_0^\infty R_{20} R_{21} r^3 dr = -\frac{1}{4a_0} (2a_0)^2 \cdot 3 \cdot 2! \cdot$$

$$\frac{a_0}{\sqrt{3}} = -3\sqrt{3} a_0^2 / \sqrt{3}. \text{ Angular: } \int Y_{0,0}^* Y_{1,0} Y_{1,0} d\Omega = \frac{1}{\sqrt{4\pi}} \cdot \frac{1}{\sqrt{3}} \text{ (3j-symbol).}$$

$$\text{Combined: } -3a_0.$$

$$W = \begin{pmatrix} 0 & \lambda \\ \lambda & 0 \end{pmatrix}, \quad \lambda = 3eE_0 a_0. \quad E^{(1)} = \pm 3eE_0 a_0. \quad \text{Good states: } |\pm\rangle =$$

$$\frac{1}{\sqrt{2}}(|200\rangle \mp |210\rangle).$$

### 31.13 L. Three Spin-1 Particles

$$H = \frac{\Delta}{\hbar^2} (\vec{S}_1 \cdot \vec{S}_2 + \vec{S}_2 \cdot \vec{S}_3 + \vec{S}_3 \cdot \vec{S}_1), \quad S_i = 1.$$

**Trick:**  $\vec{S}_1 \cdot \vec{S}_2 + \text{cyc} = \frac{1}{2}(S^2 - S_1^2 - S_2^2 - S_3^2)$ , where  $\vec{S} = \vec{S}_1 + \vec{S}_2 + \vec{S}_3$ . Each  $S_i^2 = 1(2)\hbar^2 = 2\hbar^2$ , so  $H = \frac{\Delta}{2}[S(S+1) - 6]$ .

**Spectrum** ( $1 \otimes 1 \otimes 1$ ):  $S = 0, 1, 2, 3$ .  $E_S = \frac{\Delta}{2}[S(S+1) - 6]$ :  $E_0 = -3\Delta$ ,  $E_1 = -2\Delta$ ,  $E_2 = 0$ ,  $E_3 = 3\Delta$ .

**Degeneracy:** Couple  $S_{12} = 0, 1, 2$ ; add  $S_3 = 1$ .  $S = 3$ :  $S_{12} = 2$  only  $\rightarrow 7$ .  $S = 2$ :  $S_{12} = 1, 2 \rightarrow 2 \times 5 = 10$ .  $S = 1$ :  $S_{12} = 0, 1, 2 \rightarrow 3 \times 3 = 9$ .  $S = 0$ :  $S_{12} = 1$  only  $\rightarrow 1$ . Total:  $7 + 10 + 9 + 1 = 27 = 3^3$ . ✓

**Antisymmetric states:** Only  $|S = 0, M = 0\rangle$  is totally antisymmetric (unique). Proof: For  $M = 0$  the only candidate is antisymmetrisation of  $|1, 0, -1\rangle$ :  $|A\rangle = \frac{1}{\sqrt{6}} \sum_{\sigma \in S_3} \text{sgn}(\sigma) \sigma |1, 0, -1\rangle$ . Any other  $M$  requires repeated indices  $\implies$  antisymmetrisation vanishes. Since  $\vec{S}$  is symmetric, the antisymmetric irrep must have  $S = 0$ .

### 31.14 M. Paschen-Back Effect

$$V = A \vec{L} \cdot \vec{S} + \mu(\vec{L} + 2\vec{S}) \cdot \vec{B}. \text{ Basis } |m_\ell, m_s\rangle \text{ with } m = m_\ell + m_s \text{ fixed.}$$

(i)  $A = 0$ : Choose  $\vec{B} = B\hat{z}$ .  $V = \mu B(m_\ell + 2m_s)$ .  $\Delta E = \mu B(m + m_s)$ .

(ii) **Matrix elements of  $\vec{L} \cdot \vec{S}$ :**  $\vec{L} \cdot \vec{S} = L_z S_z + \frac{1}{2}(L_+ S_- + L_- S_+)$ . Diagonal:  $\langle m_\ell, m_s | L_z S_z | m_\ell, m_s \rangle = \hbar^2 m_\ell m_s$ . Off-diagonal ( $s = \frac{1}{2}$ ):

$$\left\langle m + \frac{1}{2}, -\frac{1}{2} \right| L_+ S_- \left| m - \frac{1}{2}, \frac{1}{2} \right\rangle = \hbar^2 \sqrt{(\ell - m + \frac{1}{2})(\ell + m + \frac{1}{2})}.$$

(iii) **Stretched state**  $|\ell, \frac{1}{2}\rangle$ : no off-diagonal partner.  $\Delta E = \frac{A\hbar^2 \ell}{2} + \mu B(\ell + \frac{1}{2})$ .

(iv) **General  $m$ :** Diagonalise  $2 \times 2$  in  $\left\{ \left| m - \frac{1}{2}, \frac{1}{2} \right\rangle, \left| m + \frac{1}{2}, -\frac{1}{2} \right\rangle \right\}$ :

$$a = \frac{A\hbar^2}{2} (m - \frac{1}{2}) + \mu B(m + \frac{1}{2}),$$

$$V_m = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad c = -\frac{A\hbar^2}{2} (m + \frac{1}{2}) + \mu B(m - \frac{1}{2}),$$

$$b = \frac{A\hbar^2}{2} \sqrt{(\ell - m + \frac{1}{2})(\ell + m + \frac{1}{2})}.$$

$$\Delta E_{\pm} = \frac{1}{2} [(a+c) \pm \sqrt{(a-c)^2 + 4b^2}].$$

### 31.15 N. Low-Energy Born & Coherent Scatterers

**Born** ( $k \rightarrow 0$ ):  $e^{iq \cdot \mathbf{r}} \rightarrow 1$ .  $f(k \rightarrow 0) = -\frac{m}{2\pi\hbar^2} \int V(\mathbf{r}) d^3r$  (isotropic, angle-independent).

**Soft sphere** ( $V = V_0, r \leq a$ ):  $f = -\frac{m}{2\pi\hbar^2} \cdot \frac{4\pi a^3}{3} V_0$ .  $\sigma = 4\pi|f|^2 = \frac{16m^2 a^6 V_0^2}{9\hbar^4}$ .

$N$  scatterers at  $\mathbf{r}_i$  in box  $L$ :  $V_{\text{tot}} = \sum_{i=1}^N V(\mathbf{r} - \mathbf{r}_i)$ . Born:  $f_{\text{tot}} = f_0 \sum_i e^{iq \cdot \mathbf{r}_i}$ .

At  $k \rightarrow 0$ :  $\mathbf{q} \rightarrow 0$  so  $e^{iq \cdot \mathbf{r}_i} \rightarrow 1$ .  $f_{\text{tot}} = N f_0$ .  $\sigma = N^2 \sigma_0$  (coherent addition of amplitudes).

**Validity:**  $ka \ll 1$  (single scatterer);  $kL \ll 1$  ( $N$ -scatterer ensemble). Physics:  $\lambda \gg L$  so wave cannot resolve positions  $\implies$  amplitudes add in phase.

### 31.16 O. Wigner-Eckart Proportionality & Scalar Operator

**Problem.** Two rank- $\lambda$  ITO's  $A_\mu^{(\lambda)}, B_\mu^{(\lambda)}$ . Show  $\langle jm | A | jm' \rangle \propto \langle jm | B | jm' \rangle$  within a fixed irrep  $V(j)$ .

**Proof.** Wigner-Eckart:  $\langle jm | A_\mu^{(\lambda)} | jm' \rangle = (-)^{j-m} \frac{\langle j || A^{(\lambda)} || j \rangle}{\sqrt{2j+1}} \langle jm | \lambda \mu | jm' \rangle$ .

Same CG factor appears for  $B$ . Ratio =  $\langle j || A || j \rangle / \langle j || B || j \rangle$  is  $\mu, m, m'$ -independent. ✓

**Scalar construction.**  $S = \sum_\mu (-)^{\mu} A_\mu^{(\lambda)} B_{-\mu}^{(\lambda)}$ . Under rotation  $R$ :  $A_\mu'^{(\lambda)} = \sum_{\mu'} D_{\mu'\mu}^{(\lambda)} A_\mu^{(\lambda)}$ , same for  $B$ .

$$S' = \sum_{\mu, \mu'} (-)^{\mu} D_{\mu'\mu}^{(\lambda)} D_{\mu, -\mu'}^{(\lambda)} A_{\mu'} B_{\mu}.$$

Use  $D_{\mu''-\mu}^{(\lambda)} = (-)^{\mu''-\mu} D_{-\mu''\mu}^{(\lambda)*}$  and unitarity  $\sum_\mu D_{\mu'\mu}^{(\lambda)} D_{-\mu''\mu}^{(\lambda)*} = \delta_{\mu', -\mu''}$ :  $S' = \sum_{\mu'} (-)^{\mu'} A_{\mu'} B_{-\mu'} = S$ .  $[S, R] = 0 \forall R$ . ✓

### 31.17 P. Particle on Ring + Electric Field / Molecular Rotation

$H_0 = \frac{L^2}{2I}$ ,  $I = ma^2$ . States:  $|m\rangle = \frac{e^{im\phi}}{\sqrt{2\pi}}$ ,  $E_m^{(0)} = \frac{\hbar^2 m^2}{2I}$ . All levels doubly degenerate ( $\pm m$ ) except  $m = 0$ .

**Selection rule:**  $\langle m' | e^{i\mu\phi} | m \rangle = \delta_{m', m+\mu}$ . Perturbation  $V = \alpha \cos(n\phi) = \frac{\alpha}{2}(e^{in\phi} + e^{-in\phi}) \implies \Delta m = \pm n$ .

**E-field** ( $n = 1$ ):  $H_1 = -qUa \cos \phi$ .  $\Delta m = \pm 1$ . G.S. ( $m = 0$ ):  $E_0^{(1)} = \langle 0 | \cos \phi | 0 \rangle = 0$ . Dipole  $D_x = qa \langle \Psi | \cos \phi | \Psi \rangle = 0$  for any  $|m\rangle$  (off-diagonal). With PT:  $|\Psi_0\rangle = |0\rangle + 2\pi I q U a [ |1\rangle + |-1\rangle ]$ ; 1st-excited ( $m = \pm 1$ ): degenerate  $\Rightarrow$  diag  $2 \times 2$ :  $V = \frac{-qUa}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ ;  $E_{\pm}^{(1)} = \mp \frac{qUa}{2}$ ; good states  $\frac{1}{\sqrt{2}}(|1\rangle \pm |-1\rangle)$ .

**Ethane** ( $n = 3$ ):  $H_1 = W \cos 3\phi$ .  $\Delta m = \pm 3$ . G.S.:  $E_0^{(1)} = 0$ .  $|\Psi_0\rangle = |0\rangle - \frac{2\pi IW}{9\hbar^2} [|3\rangle + |-3\rangle]$ . (Only  $m = \pm 3$  couple to  $m = 0$ ; gap  $\propto 9$ .)

### 31.18 Q. Symmetry of $J=2j$ Multiplet ( $j \otimes j$ )

**Claim:** In  $j \otimes j$ , the  $J = 2j$  multiplet is symmetric and  $J = 2j - 1$  is antisymmetric under particle exchange.

**Proof.** *Symmetric:*  $|J=2j, M=2j\rangle = |j, j\rangle \otimes |j, j\rangle$  is manifestly symmetric.  $J_- = J_{1-} + J_{2-}$  is symmetric  $\Rightarrow$  all  $|2j, M\rangle$  symmetric.

*Antisymmetric:*  $|2j-1, 2j-1\rangle = \alpha |j, j-1; j, j\rangle + \beta |j, j; j, j-1\rangle$ . Apply  $J_+$ :  $J_+ |2j-1, 2j-1\rangle = 0 \Rightarrow (\alpha + \beta)\sqrt{2j} |j, j; j, j\rangle = 0 \Rightarrow \beta = -\alpha$ .  $\therefore$  state  $\propto |j, j-1; j, j\rangle - |j, j; j, j-1\rangle$ : antisymmetric. Symmetric  $J_-$  preserves this.

**General rule (CG symmetry):**  $\langle j m_1 j m_2 | J M \rangle = (-1)^{2j-J} \langle j m_2 j m_1 | J M \rangle$ . Symmetric iff  $2j - J$  even; antisymmetric iff  $2j - J$  odd.

**Application (He,  $n=2, l=1$ ):**  $l_1=l_2=1$ :  $L = 0$  (sym),  $L = 1$  (antisym),  $L = 2$  (sym). Fermions need total antisymmetry  $\Rightarrow$  spin antisym for sym  $L$  and vice versa.  $L=1$  (antisym spatial) needs  $S=1$  (sym spin). OK.  $L=0$  (sym) needs  $S=0$  (antisym).  $J = 0$  only.  $L=0, S=1$ : both symmetric  $\Rightarrow$  **forbidden** for fermions.

### 31.19 R. Two-Level System: Exact Rabi (Delta Pulse)

$H_0 |a\rangle = E_a |a\rangle$ ,  $H_0 |b\rangle = E_b |b\rangle$ .  $V(t) = [|b\rangle \langle a| + |a\rangle \langle b|] \delta_{\Delta}(t)$ ,  $\delta_{\Delta} = 1/\Delta$  for  $0 \leq t \leq \Delta$ , else 0.

**1st-order PT** ( $\delta_{\Delta} \rightarrow \delta$ ):  $c_b = -\frac{i\alpha}{\hbar}$ ,  $P_b = |\alpha/\hbar|^2$ . **2nd-order:**  $c_b^{(2)} = 0$  (no intermediate path back).

**Exact** ( $|\omega_{ba}\Delta| \ll 1$ ): Coupled eqs for  $0 < t < \Delta$ :  $\dot{c}_a = -\frac{i\alpha^*}{\hbar\Delta} c_b$ ,  $\dot{c}_b = -\frac{i\alpha}{\hbar\Delta} c_a$ .  $\ddot{c}_a = -\frac{|\alpha|^2}{\hbar^2\Delta^2} c_a$ . Solution with  $c_a(0) = 1$ :

$$c_a(t) = \cos \frac{|\alpha|t}{\hbar\Delta}, \quad c_b(t) = -\frac{i\alpha}{|\alpha|} \sin \frac{|\alpha|t}{\hbar\Delta}.$$

$$\text{At } t = \Delta: \quad \boxed{P_a = \cos^2 \frac{|\alpha|}{\hbar}, \quad P_b = \sin^2 \frac{|\alpha|}{\hbar}}.$$

For  $|\alpha| \ll \hbar$ :  $\sin^2(|\alpha|/\hbar) \approx |\alpha|^2/\hbar^2$  recovers 1st-order PT. Full Rabi oscillation when  $|\alpha| = \pi\hbar/2$  ( $\pi$ -pulse).

### 31.20 S. Detailed Balance in TDPT

**Claim:**  $P_{m \rightarrow n}^{(1)}(t) = P_{n \rightarrow m}^{(1)}(t)$ .

**Proof.**

$$P_{m \rightarrow n}^{(1)} = \frac{1}{\hbar^2} \left| \int_{t_0}^t V_{nm}(t') e^{i\omega_{nm}t'} dt' \right|^2 = \frac{1}{\hbar^2} \left| \int_{t_0}^t V_{mn}^*(t') e^{-i\omega_{mn}t'} dt' \right|^2.$$

Taking the complex conjugate of the integral:  $\left| \int V_{mn}^* e^{-i\omega_{mn}t'} dt' \right|^2 = \left| \int V_{mn} e^{i\omega_{mn}t'} dt' \right|^2 = P_{n \rightarrow m}^{(1)}$ .  $\checkmark$

Key: uses  $V_{nm} = V_{mn}^*$  (Hermiticity) and  $\omega_{nm} = -\omega_{mn}$ .  $|z|^2 = |z^*|^2$  does the rest.

## 32 Solved: Variational Method

### 32.1 1D Linear Potential ( $V = \lambda|x|$ )

**Trial:** Gaussian  $\psi(x) = \left(\frac{\alpha}{\pi}\right)^{1/4} e^{-\alpha x^2/2}$ . **Kinetic:**  $\langle T \rangle = \frac{\hbar^2 \alpha}{4m}$  (Standard Gaussian result). **Potential:**  $\langle V \rangle = \lambda \sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{\infty} |x| e^{-\alpha x^2} dx = 2\lambda \sqrt{\frac{\alpha}{\pi}} \int_0^{\infty} x e^{-\alpha x^2} dx$ .

Using  $\int_0^{\infty} x e^{-\alpha x^2} dx = \frac{1}{2\alpha}$ :

$$\langle V \rangle = \frac{\lambda}{\sqrt{\pi\alpha}}$$

**Total Energy:**  $E(\alpha) = \frac{\hbar^2 \alpha}{4m} + \frac{\lambda}{\sqrt{\pi\alpha}}$ . **Minimize:**  $\frac{dE}{d\alpha} = \frac{\hbar^2}{4m} - \frac{\lambda}{2\sqrt{\pi}} \alpha^{-3/2} = 0$ .

$$\alpha^{3/2} = \frac{2m\lambda}{\hbar^2 \sqrt{\pi}} \Rightarrow \alpha_0 = \left( \frac{2m\lambda}{\hbar^2 \sqrt{\pi}} \right)^{2/3}$$

Substitute back for  $E_{\min}$ .

### 32.2 Helium Atom (Screening)

$H = -\frac{1}{2}\nabla_1^2 - \frac{1}{2}\nabla_2^2 - \frac{Z}{r_1} - \frac{Z}{r_2} + \frac{1}{|r_1 - r_2|}$ . **Trial:**  $\Psi = \psi_{1s}(Z^*, r_1)\psi_{1s}(Z^*, r_2)$  (Uncorrelated product). **Virial Scaling:** For hydrogenic state with  $Z^*$ :  $\langle T \rangle = (Z^*)^2$ ,  $\langle V_{nuc} \rangle = -2ZZ^*$ . **Interaction:**  $\langle \frac{1}{r_{12}} \rangle = \frac{5}{8}Z^*$  (Standard Integral). **Total:**  $E(Z^*) = 2(ZZ^*)^2 - 4ZZ^* + \frac{5}{8}Z^*$ . **Minimize:**  $\frac{dE}{dZ^*} = 4Z^* - 4Z + \frac{5}{8} = 0 \Rightarrow Z^* = Z - \frac{5}{16}$ . **Result:**  $E_{var} = -2(Z - \frac{5}{16})^2$  Ry. *Physics:* The other electron screens 5/16 of the nuclear charge.

### 32.3 Deuteron (Square Well)

Trial  $\psi(r) = e^{-\alpha r}$  for attractive well  $V = -V_0$  ( $r < a$ ).  $\langle H \rangle = \frac{\hbar^2 \alpha^2}{2\mu} - V_0[1 - e^{-2\alpha a}(1 + 2\alpha a)]$ . Minimizing gives bound on  $V_0$  for bound state existence.

## 33 Identical Particles in HO

Two electrons in 1D HO.  $V_{int} = \lambda\delta(x_1 - x_2)$ . Perturbation theory. **Basis:** Unperturbed states  $|n_1, n_2\rangle$ . Ground state ( $n_1 = 0, n_2 = 0$ ): Spatial part  $\psi_0(x_1)\psi_0(x_2)$  is symmetric. Spin must be Antisymmetric (Singlet  $S = 0$ ). Shift:  $\Delta E = \lambda \int |\psi_0(x_1)|^2 |\psi_0(x_2)|^2 \delta(x_1 - x_2) dx_1 dx_2$ .  $\Delta E = \lambda \int |\psi_0(x)|^4 dx = \lambda \left(\frac{m\omega}{\pi\hbar}\right) \int e^{-2m\omega x^2/\hbar} dx$ . **Result (Singlet):**  $\Delta E = \lambda \sqrt{\frac{m\omega}{2\pi\hbar}}$ .

**Excited State** ( $n_1 = 0, n_2 = 1$ ): 1. **Triplet** ( $S = 1$ ): Spatial Antisym  $\psi_A = \frac{1}{\sqrt{2}}(\psi_0(1)\psi_1(2) - \psi_1(1)\psi_0(2))$ .  $\psi_A(x, x) = 0$  (Pauli exclusion). **Result:**  $\Delta E_{trip} = \langle \psi_A | \delta | \psi_A \rangle = 0$ .

2. **Singlet** ( $S = 0$ ): Spatial Sym  $\psi_S = \frac{1}{\sqrt{2}}(\psi_0(1)\psi_1(2) + \psi_1(1)\psi_0(2))$ .  $\Delta E_{sing} = \lambda \int |\psi_S(x, x)|^2 dx = \lambda \cdot 2 \int |\psi_0(x)\psi_1(x)|^2 dx$ .

## 34 Solved: Identical Particles

### 34.1 Exchange Energy (He excited states)

Two electrons in states  $\psi_a, \psi_b$ . Interaction  $V(r_1, r_2)$ . **Direct Integral:**  $J = \int |\psi_a(1)|^2 |\psi_b(2)|^2 V d\tau$ . **Exchange Integral:**  $K = \int \psi_a^*(1)\psi_b^*(2)V\psi_a(2)\psi_b(1) d\tau$ . **Energies:**

- **Singlet** ( $S = 0$ ): Spatial Sym  $\Rightarrow E_S = E_a + E_b + J + K$ .
- **Triplet** ( $S = 1$ ): Spatial Anti  $\Rightarrow E_T = E_a + E_b + J - K$ .

Since  $K > 0$  usually (Coulomb repulsion),  $E_T < E_S$  (Hund's Rule). **Physical reason:** Triplet spatial wavefunction vanishes at  $r_1 = r_2$ , reducing repulsion.

### 34.2 Interacting Bosons

Two Bosons in 1D HO.  $V = \lambda\delta(x_1 - x_2)$ . States  $|n_1, n_2\rangle$ . **Case 1: Ground State**  $|0, 0\rangle$ .  $\Delta E = \lambda \int |\psi_0(x)|^4 dx = \lambda \sqrt{\frac{m\omega}{2\pi\hbar}}$ . **Case 2: First Excited**  $|1, 0\rangle$ . Symmetrized state:  $\Psi = \frac{1}{\sqrt{2}}[\psi_0(1)\psi_1(2) + \psi_1(1)\psi_0(2)]$ .  $\Delta E = \lambda \int |\Psi(x, x)|^2 dx = \lambda \cdot 2 \int |\psi_0(x)|^2 |\psi_1(x)|^2 dx$ .  $\Delta E = \lambda \sqrt{\frac{m\omega}{2\pi\hbar}}$ . (Same as GS! due to Gaussian properties). *Note:* If Fermions,  $\Psi_A \propto [\psi_0(1)\psi_1(2) - \dots]$ .  $\Psi(x, x) = 0 \Rightarrow \Delta E = 0$ .

## 35 Gaussian Pulse

1D HO in ground state  $|0\rangle$ . Perturbation  $H'(t) = -qExe^{-t^2/\tau^2}$ . Find prob to transition to  $|1\rangle$  at  $t \rightarrow \infty$ .  $c_1(t) = -\frac{i}{\hbar} \int_{-\infty}^t e^{i\omega_{10}t'} V_{10}(t') dt'$ .  $V_{10} = -qE \langle 1 | x | 0 \rangle e^{-t^2/\tau^2} = -qE \ell_0 e^{-t^2/\tau^2}$ .  $\text{integral } I = \int_{-\infty}^{\infty} e^{-t^2/\tau^2 + i\omega t} dt = \sqrt{\pi\tau^2} e^{-\omega^2\tau^2/4}$ .  $c_1(\infty) = -\frac{i}{\hbar} (-qE \ell_0) \sqrt{\pi\tau^2} e^{-\omega^2\tau^2/4}$ . **Result:**  $P_{0 \rightarrow 1} = \frac{\pi q^2 E^2 \ell_0^2 \tau^2}{\hbar^2} e^{-\omega^2\tau^2/2}$ . *Note:* If pulse is very slow ( $\tau \rightarrow \infty$ ),  $P \rightarrow 0$  (Adiabatic).

## 36 Solved: Time-Dependent Dynamics

### 36.1 Driven Two-Level System (Resonance)

$H_0 = \begin{pmatrix} E_a & 0 \\ 0 & E_b \end{pmatrix}$ ,  $V(t) = \gamma \cos(\omega t) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ . Let  $\omega_0 = (E_b - E_a)/\hbar$ . Detuning  $\Delta = \omega - \omega_0$ . **Perturbation Limit:** If  $\gamma \ll \hbar\Delta$  (Weak/Far detuned):  $P(t) \approx \frac{\gamma^2}{\hbar^2 \Delta^2} \sin^2(\Delta t/2)$ . (Matches 1st order TDPT result).

## 37 Solved: Spin Archetypes

### 37.1 Hyperfine Splitting (H-Ground State)

$H_{hf} = \mathbf{A} \mathbf{S}_e \cdot \mathbf{S}_p$ . ( $e$  and  $p$  are spin-1/2). Total spin  $\mathbf{F} = \mathbf{S}_e + \mathbf{S}_p$ . Identity:  $\mathbf{S}_e \cdot \mathbf{S}_p = \frac{1}{2}(\mathbf{F}^2 - \mathbf{S}_e^2 - \mathbf{S}_p^2)$ . Eigenvalues:  $\frac{\hbar^2}{2}[F(F+1) - 3/4 - 3/4]$ . **Triplet** ( $F = 1$ ):  $E_T = \frac{\hbar^2 A}{2}[2 - 3/2] = \frac{\hbar^2 A}{4}$ . **Singlet** ( $F = 0$ ):  $E_S = \frac{\hbar^2 A}{2}[0 - 3/2] = -\frac{\hbar^2 A}{4}$ .

$-\frac{3\hbar^2 A}{4}$ . **Splitting:**  $\Delta E = \hbar^2 A$  (21 cm line). Degeneracy: Triplet (3), Singlet (1).

### 37.2 Spin Precession (Exact)

$H = -\gamma B S_z$ . Initial state  $|+\rangle_x = \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$ . Time evolution:  $|\psi(t)\rangle = \frac{1}{\sqrt{2}}(e^{-iE_1 t/\hbar} |\uparrow\rangle + e^{-iE_1 t/\hbar} |\downarrow\rangle)$ .  $E_{\uparrow/\downarrow} = \mp \hbar \omega_L/2$ .  $|\psi(t)\rangle = \frac{1}{\sqrt{2}}(e^{i\omega_L t/2} |\uparrow\rangle + e^{-i\omega_L t/2} |\downarrow\rangle)$ . Spin vector  $\langle \mathbf{S} \rangle(t)$ :  $\langle S_x \rangle = \frac{\hbar}{2} \cos(\omega_L t)$ ,  $\langle S_y \rangle = \frac{\hbar}{2} \sin(\omega_L t)$ . Precesses about  $z$  with Larmor freq  $\omega_L = \gamma B$ .

## 38 Solved: Virial Theorem Tricks

**Theorem:**  $2\langle T \rangle = \langle \mathbf{r} \cdot \nabla V \rangle$ . For potentials  $V(r) = \alpha r^n$ :  $2\langle T \rangle = n\langle V \rangle$ .

### 38.1 Harmonic Oscillator ( $n = 2$ )

$2\langle T \rangle = 2\langle V \rangle \implies \langle T \rangle = \langle V \rangle = E_n/2$ . Useful for perturbations like  $H' = \lambda x^2$ .  $E^{(1)} = \lambda \langle x^2 \rangle = \lambda \frac{2\langle V_0 \rangle}{m\omega^2} = \lambda \frac{E_n}{m\omega^2}$ .

### 38.2 Hydrogen / Coulomb ( $n = -1$ )

$2\langle T \rangle = -\langle V \rangle$ . Total  $E = \langle T \rangle + \langle V \rangle = -\langle T \rangle = \frac{1}{2}\langle V \rangle$ . Allows instant calculation of  $\langle 1/r \rangle$ :  $\langle \frac{1}{r} \rangle = -\frac{1}{e^2} \langle V \rangle = -\frac{2E_n}{e^2} = \frac{2}{e^2} \frac{13.6 Z^2}{n^2}$ .

## 39 Mathematical Tools

### 39.1 Cylindrical

$$\begin{aligned}\nabla f &= \hat{\rho} \frac{\partial f}{\partial \rho} + \hat{\phi} \frac{1}{\rho} \frac{\partial f}{\partial \phi} + \hat{z} \frac{\partial f}{\partial z} \\ \nabla \cdot \mathbf{A} &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z} \\ \nabla \times \mathbf{A} &= \left( \frac{1}{\rho} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) \hat{\rho} + \left( \frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \right) \hat{\phi} + \frac{1}{\rho} \left( \frac{\partial}{\partial \rho} (\rho A_\phi) - \frac{\partial A_\rho}{\partial \phi} \right) \hat{z} \\ \nabla^2 f &= \frac{\partial^2 f}{\partial \rho^2} + \frac{1}{\rho} \frac{\partial f}{\partial \rho} + \frac{1}{\rho^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2} \\ dV &= \rho d\rho d\phi dz\end{aligned}$$

### 39.2 Spherical

$$\begin{aligned}\nabla f &= \hat{r} \frac{\partial f}{\partial r} + \hat{\theta} \frac{1}{r} \frac{\partial f}{\partial \theta} + \hat{\phi} \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \\ \nabla \cdot \mathbf{A} &= \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi} \\ \nabla \times \mathbf{A} &= \frac{1}{r \sin \theta} \left( \frac{\partial}{\partial \theta} (\sin \theta A_\phi) - \frac{\partial A_\phi}{\partial \theta} \right) \hat{r} + \frac{1}{r} \left( \frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r} (r A_\phi) \right) \hat{\theta} + \frac{1}{r} \left( \frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right) \hat{\phi} \\ \nabla^2 f &= \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2} \\ \nabla^2 f &= \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial f}{\partial r} \right) - \frac{L^2}{\hbar^2 r^2} f \\ L^2 &= -\hbar^2 \left[ \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right] \\ dV &= r^2 \sin \theta dr d\theta d\phi\end{aligned}$$

### 39.3 Vector Identities

$$\begin{aligned}\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) &= \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) \\ (\mathbf{A} \times \mathbf{B}) \times \mathbf{C} &= (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{B} \cdot \mathbf{C})\mathbf{A} \\ \cos a \cos b &= \frac{1}{2} [\cos(a-b) + \cos(a+b)] \\ \sin a \sin b &= \frac{1}{2} [\cos(a-b) - \cos(a+b)] \\ \sin a \cos b &= \frac{1}{2} [\sin(a-b) + \sin(a+b)]\end{aligned}$$

### 39.4 Spherical harmonics in $x, y, z$

$$\begin{aligned}x &= \sqrt{\frac{2\pi}{3}} r (Y_1^1 + Y_1^{-1}), \quad y = i\sqrt{\frac{2\pi}{3}} r (Y_1^1 - Y_1^{-1}), \\ z &= 2r\sqrt{\frac{3}{4\pi}} Y_1^0, \quad x^2 - y^2 = 2\sqrt{\frac{2\pi}{15}} r^2 (Y_2^2 + Y_2^{-2}), \\ x^2 + y^2 &= \frac{4\sqrt{\pi}}{3} r^2 \left( Y_0^0 - \frac{1}{\sqrt{5}} Y_2^0 \right), \quad z^2 = \frac{2\sqrt{\pi}}{3} r^2 \left( Y_0^0 + \frac{2}{\sqrt{5}} Y_2^0 \right).\end{aligned}$$

### Angular Momentum Commutators

$$\begin{aligned}[J_i, J_j] &= i\hbar \epsilon_{ijk} J_k & [J^2, J_i] &= 0 & [L_i, S_j] &= 0 \\ [J_z, J_\pm] &= \pm \hbar J_\pm & [J_+, J_-] &= 2\hbar J_z & [J^2, J_\pm] &= 0 \\ J_\mp J_\pm &= J^2 - J_z^2 \mp \hbar J_z & \mathbf{L} \times \mathbf{L} &= i\hbar \mathbf{L} & \mathbf{J} &= \mathbf{L} + \mathbf{S}\end{aligned}$$

| Operator             | Action on $ j, m\rangle$                                 | $\Delta j$ ? |
|----------------------|----------------------------------------------------------|--------------|
| $J_z$                | $m\hbar  j, m\rangle$                                    | no           |
| $J_\pm$              | $\hbar\sqrt{j(j+1) - m(m \pm 1)}  j, m \pm 1\rangle$     | no           |
| $J^2$                | $j(j+1)\hbar^2  j, m\rangle$                             | no           |
| E1 (electric dipole) | mixes $j \rightarrow j, j \pm 1$ (no $0 \rightarrow 0$ ) | yes          |
| M1 (magnetic dipole) | mixes $j \rightarrow j, j \pm 1$ (no $0 \rightarrow 0$ ) | yes          |
| E2 (quadrupole)      | mixes $j \rightarrow j, j \pm 1, \pm 2$ (restricted)     | yes          |

### Spin-Orbit & Conservation ( $H_{SO} \propto \mathbf{L} \cdot \mathbf{S}$ )

$$\begin{aligned}[\mathbf{L} \cdot \mathbf{S}, \mathbf{J}] &= 0 & [\mathbf{L} \cdot \mathbf{S}, L^2] &= 0 & [\mathbf{L} \cdot \mathbf{S}, S^2] &= 0 \\ [\mathbf{L} \cdot \mathbf{S}, L_z] &\neq 0 & [\mathbf{L} \cdot \mathbf{S}, S_z] &\neq 0 & \mathbf{L} \cdot \mathbf{S} &= \frac{1}{2}(J^2 - L^2 - S^2)\end{aligned}$$

## 40 Integrals

$$\begin{aligned}\int_0^\infty r^n e^{-ar} dr &= \frac{n!}{a^{n+1}} \\ \int_0^\infty e^{-\mu r} \sin(qr) dr &= \frac{q}{\mu^2 + q^2} \quad \int_{-\infty}^\infty e^{-ax^2 + bx} dx = \sqrt{\frac{\pi}{a}} e^{b^2/4a} \\ \int_0^L x \sin\left(\frac{n\pi x}{L}\right) dx &= \frac{(-1)^{n+1} L^2}{n\pi} \\ \int_0^L x^2 \sin^2\left(\frac{n\pi x}{L}\right) dx &= L^3 \left( \frac{1}{3} - \frac{1}{2n^2 \pi^2} \right) \\ \int_0^L x \cos\left(\frac{n\pi x}{L}\right) dx &= \frac{L^2}{n^2 \pi^2} [(-1)^n - 1] \\ \int \frac{dx}{a^2 - x^2} &= \frac{1}{2a} \ln \left| \frac{x+a}{x-a} \right| + C. \quad \int_{-\infty}^\infty x^2 e^{-ax^2} dx = \frac{1}{2} \sqrt{\frac{\pi}{a^3}} \\ \int \frac{e^{-\mu r}}{r} e^{i\mathbf{q} \cdot \mathbf{r}} d^3 r &= \frac{4\pi}{q^2 + \mu^2} \quad (\text{Coulomb: } \mu \rightarrow 0 \Rightarrow 4\pi/q^2)\end{aligned}$$

For a scalar function  $u(r, \theta, \phi)$  that depends exclusively on the radial coordinate  $r$ :

$$\begin{aligned}\int_V u(r) dV &= \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta \int r^2 u(r) dr \\ &= 4\pi \int r^2 u(r) dr\end{aligned}$$

### Useful Taylor Series ( $|x| \ll 1$ ):

$$\begin{aligned}e^x &\approx 1 + x + \frac{x^2}{2!} + \dots & \sin x &\approx x - \frac{x^3}{3!} + \dots & \cos x &\approx 1 - \frac{x^2}{2!} + \dots \\ (1+x)^n &\approx 1 + nx + \frac{n(n-1)}{2} x^2 & \ln(1+x) &\approx x - \frac{x^2}{2} + \dots & \frac{1}{1 \mp x} &\approx 1 \pm x + x^2\end{aligned}$$

### 40.1 22.2 Delta Function

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^\infty e^{ikx} dk = \lim_{\epsilon \rightarrow 0} \frac{1}{\pi} \frac{\epsilon}{\epsilon^2 + x^2} = \lim_{a \rightarrow 0} \frac{e^{-x^2/a^2}}{a\sqrt{\pi}}$$

$$\delta(ax) = \frac{1}{|a|} \delta(x). \quad \delta(g(x)) = \sum_i \frac{\delta(x-x_i)}{|g'(x_i)|} \quad (g(x_i) = 0).$$

$$\int \delta(x-a) f(x) dx = f(a)$$

$$Y_1^0 = \sqrt{\frac{3}{4\pi}} \frac{z}{r}, \quad Y_1^{\pm 1} = \mp \sqrt{\frac{3}{8\pi}} \frac{x \pm iy}{r}$$

### 40.2 CG Coefficients: $\frac{1}{2} \otimes \frac{1}{2}$

$$|l_1, l_2; m_1 m_2\rangle \leftrightarrow |l, m\rangle.$$

From product to coupled states:

$$\begin{pmatrix} |0,0\rangle \\ |1,1\rangle \\ |1,0\rangle \\ |1,-1\rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & -1 & 0 \\ \sqrt{2} & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} |\uparrow\uparrow\rangle \\ |\uparrow\downarrow\rangle \\ |\downarrow\uparrow\rangle \\ |\downarrow\downarrow\rangle \end{pmatrix}.$$

From coupled to product states:

$$\begin{pmatrix} |\uparrow\uparrow\rangle \\ |\uparrow\downarrow\rangle \\ |\downarrow\uparrow\rangle \\ |\downarrow\downarrow\rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & \sqrt{2} & 0 & 0 \\ 1 & 0 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} |0,0\rangle \\ |1,1\rangle \\ |1,0\rangle \\ |1,-1\rangle \end{pmatrix}.$$

## 1. Gaussian Integrals

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-ax^2} dx &= \sqrt{\frac{\pi}{a}}, & \int_{-\infty}^{\infty} x^2 e^{-ax^2} dx &= \frac{1}{2} \sqrt{\frac{\pi}{a^3}} \\ \int_{-\infty}^{\infty} e^{-ax^2+bx} dx &= \sqrt{\frac{\pi}{a}} e^{b^2/4a}, & \int_0^{\infty} e^{-ax^2} dx &= \frac{1}{2} \sqrt{\frac{\pi}{a}} \\ \int_{-\infty}^{\infty} x^{2n} e^{-ax^2} dx &= \frac{(2n)! \sqrt{\pi}}{n! 4^n a^{n+\frac{1}{2}}}, & \int_{-\infty}^{\infty} x^{2n+1} e^{-ax^2} dx &= 0 \\ \int_0^{\infty} x^{2n} e^{-ax^2} dx &= \frac{(2n-1)!! \sqrt{\pi}}{2^{n+1} a^{n+\frac{1}{2}}}, & \int_0^{\infty} x^{2n+1} e^{-ax^2} dx &= \frac{n!}{2a^{n+1}}\end{aligned}$$

## 2. Gamma & Beta

$$\begin{aligned}\int_0^{\infty} x^n e^{-ax} dx &= \frac{n!}{a^{n+1}}, & \int_0^{\infty} x^{\nu-1} e^{-ax} dx &= \frac{\Gamma(\nu)}{a^\nu} \\ \Gamma(n+1) &= n!, & \Gamma\left(\frac{1}{2}\right) &= \sqrt{\pi} \\ \Gamma\left(n+\frac{1}{2}\right) &= \frac{(2n-1)!!}{2^n} \sqrt{\pi} \\ \int_0^1 x^m (1-x)^n dx &= B(m+1, n+1) = \frac{m! n!}{(m+n+1)!}\end{aligned}$$

## 3. Harmonic Oscillator

$$\begin{aligned}\int_{-\infty}^{\infty} H_m H_n e^{-x^2} dx &= \sqrt{\pi} 2^n n! \delta_{mn} \\ \langle n|x|m\rangle &= \sqrt{\frac{\hbar}{2M\omega}} (\sqrt{m} \delta_{n,m-1} + \sqrt{m+1} \delta_{n,m+1}) \\ \langle n|x^2|m\rangle &= \frac{\hbar}{2M\omega} [\sqrt{m(m-1)} \delta_{n,m-2} + (2m+1) \delta_{nm} \\ &\quad + \sqrt{(m+1)(m+2)} \delta_{n,m+2}] \\ \langle n|p^2|m\rangle &= -\frac{M\omega\hbar}{2} [\sqrt{m(m-1)} \delta_{n,m-2} - (2m+1) \delta_{nm} \\ &\quad + \sqrt{(m+1)(m+2)} \delta_{n,m+2}]\end{aligned}$$

## 4. Spherical / Angular

$$\begin{aligned}\int Y_{\ell'}^{m'} * Y_{\ell}^m d\Omega &= \delta_{\ell'\ell'} \delta_{mm'}, & \int d\Omega &= 4\pi \\ \int_0^{\pi} P_{\ell} P_{\ell'} \sin \theta d\theta &= \frac{2}{2\ell+1} \delta_{\ell'\ell'}, & \int_0^{\pi} e^{i\eta\phi} d\phi &= 2\pi \delta_{n0} \\ \int_0^{\pi} P_{\ell}^m P_{\ell'}^m \sin \theta d\theta &= \frac{2(\ell+m)!}{(2\ell+1)(\ell-m)!} \delta_{\ell'\ell'} \\ \int Y_{\ell_1}^{m_1} Y_{\ell_2}^{m_2} Y_{\ell_3}^{m_3} * d\Omega &= \sqrt{\frac{(2\ell_1+1)(2\ell_2+1)}{4\pi(2\ell_3+1)}} C_{\ell_1 0 \ell_2 0}^{\ell_3 0} C_{\ell_1 m_1 \ell_2 m_2}^{\ell_3 m_3} \\ \int_0^{\pi} e^{iqr \cos \theta} \sin \theta d\theta &= \frac{2 \sin(qr)}{qr} = 2 j_0(qr)\end{aligned}$$

## 5. Fourier Transforms [ $\tilde{f}(k) = \int e^{-ikx} f(x) dx$ ]

$$\begin{aligned}\mathcal{F}[e^{-a|x|}] &= \frac{2a}{a^2 + k^2}, & \mathcal{F}[e^{-ax^2}] &= \sqrt{\frac{\pi}{a}} e^{-k^2/4a} \\ \mathcal{F}[\delta(x-x_0)] &= e^{-ikx_0}, & \mathcal{F}[e^{ik_0x}] &= 2\pi \delta(k-k_0) \\ \mathcal{F}[\Theta(x)e^{-ax}] &= \frac{1}{a+ik}, & \mathcal{F}_{3D}\left[\frac{1}{r}\right] &= \frac{4\pi}{k^2}\end{aligned}$$

**Yukawa / Coulomb:**  $\int e^{i\vec{q}\cdot\vec{r}} \frac{e^{-\mu r}}{r} d^3r = \frac{4\pi}{q^2 + \mu^2} \xrightarrow{\mu \rightarrow 0} \frac{4\pi}{q^2}$

## 6. Born Approximation & Scattering

$$\begin{aligned}f(\theta) &= -\frac{m}{2\pi\hbar^2} \int e^{i\vec{q}\cdot\vec{r}} V(r) d^3r, \quad \vec{q} = \vec{k}_i - \vec{k}_f, \quad |\vec{q}| = 2k \sin \frac{\theta}{2} \\ \int_0^{\infty} V(r) j_0(qr) r^2 dr &= \int_0^{\infty} V(r) \frac{\sin(qr)}{qr} r^2 dr \\ \int_0^{\infty} j_{\ell}(kr) j_{\ell}(k'r) r^2 dr &= \frac{\pi}{2k^2} \delta(k-k') \\ \int_0^{\infty} \frac{\sin(qr)}{qr} e^{-\mu r} r^2 dr &= \frac{1}{q^2 + \mu^2} \\ \int_0^{\infty} e^{-(a+iq)r} r^n dr &= \frac{n!}{(a+iq)^{n+1}}\end{aligned}$$

## 7. Hydrogen Atom Radial

$$\begin{aligned}\int_0^{\infty} r^n e^{-ar} dr &= \frac{n!}{a^{n+1}} \\ \langle r \rangle_{n\ell} &= \frac{a_0}{2} [3n^2 - \ell(\ell+1)] \\ \langle r^2 \rangle_{n\ell} &= \frac{n^2 a_0^2}{2} [5n^2 + 1 - 3\ell(\ell+1)] \\ \langle r^{-1} \rangle_{n\ell} &= \frac{1}{n^2 a_0}, & \langle r^{-2} \rangle_{n\ell} &= \frac{2}{n^3 (2\ell+1) a_0^2} \\ \langle r^{-3} \rangle_{n\ell} &= \frac{1}{n^3 \ell(\ell+\frac{1}{2})(\ell+1) a_0^3}\end{aligned}$$

## 8. Delta Function & Principal Value

$$\begin{aligned}\delta(f(x)) &= \sum_i \frac{\delta(x-x_i)}{|f'(x_i)|}, & \delta^{(3)}(\vec{r}) &= \frac{1}{(2\pi)^3} \int e^{i\vec{k}\cdot\vec{r}} d^3k \\ \frac{1}{x \pm i\epsilon} &= \text{P.V.} \frac{1}{x} \mp i\pi \delta(x), & \nabla^2\left(\frac{1}{r}\right) &= -4\pi \delta^{(3)}(\vec{r}) \\ \delta^{(3)}(\vec{r}-\vec{r}') &= \frac{\delta(r-r')}{r^2} \frac{\delta(\theta-\theta')}{\sin \theta} \delta(\phi-\phi')\end{aligned}$$

## 9. Contour & Special

$$\begin{aligned}\int_0^{\infty} \frac{\sin(ax)}{x} dx &= \frac{\pi}{2} \text{sgn}(a), & \int_0^{\infty} \frac{1-\cos(ax)}{x^2} dx &= \frac{\pi|a|}{2} \\ \int_0^{\infty} j_0(kr) k^2 dk &= \frac{\pi}{2} \delta(r), & \oint_C \frac{e^{ikz}}{z^2+a^2} dz &= \frac{\pi}{a} e^{-a|k|} \\ \int_{-\infty}^{\infty} \frac{dx}{(x^2+a^2)^n} &= \frac{\pi (2n-2)!}{[(n-1)!]^2 4^{n-1} a^{2n-1}}\end{aligned}$$

## 10. Trigonometric Integrals

$$\begin{aligned}\int_0^{2\pi} \sin(mx) \sin(nx) dx &= \pi \delta_{mn}, & \int_0^{2\pi} \cos(mx) \cos(nx) dx &= \pi \delta_{mn} \\ \int_0^{2\pi} \sin(mx) \cos(nx) dx &= 0, & \int_0^L \sin \frac{m\pi x}{L} \sin \frac{n\pi x}{L} dx &= \frac{L}{2} \delta_{mn} \\ \int_0^{\pi} \sin(mx) \sin(nx) dx &= \frac{\pi}{2} \delta_{mn}, & \int_0^{\pi} \cos(mx) \cos(nx) dx &= \frac{\pi}{2} \delta_{mn}\end{aligned}$$

**Powers over  $[0, \pi]$  (Wallis,  $n \geq 0$ ):**

$$\begin{aligned}\int_0^{\pi} \sin^{2n} \theta d\theta &= \frac{(2n-1)!!}{(2n)!!} \pi, & \int_0^{\pi} \sin^{2n+1} \theta d\theta &= \frac{2(2n)!!}{(2n+1)!!} \\ \int_0^{2\pi} \sin^{2n} \theta d\theta &= \frac{(2n-1)!!}{(2n)!!} 2\pi, & \int_0^{\pi} \cos^{2n+1} \theta d\theta &= 0\end{aligned}$$

**Spherical angular ( $m, n \geq 0$ ):**

$$\begin{aligned}\int_0^{\pi} \cos^m \theta \sin^n \theta d\theta &= \frac{\Gamma\left(\frac{m+1}{2}\right) \Gamma\left(\frac{n+1}{2}\right)}{2\Gamma\left(\frac{m+n+2}{2}\right)} \\ \int_0^{\pi} \cos^{2n} \theta \sin \theta d\theta &= \frac{2}{2n+1}, & \int_0^{\pi} \cos^{2n+1} \theta \sin \theta d\theta &= 0\end{aligned}$$

**Gaussian-trig:**

$$\begin{aligned}\int_{-\infty}^{\infty} e^{-ax^2} \cos(bx) dx &= \sqrt{\frac{\pi}{a}} e^{-b^2/4a} \\ \int_{-\infty}^{\infty} x e^{-ax^2} \sin(bx) dx &= \frac{b}{2a} \sqrt{\frac{\pi}{a}} e^{-b^2/4a} \\ \int_{-\infty}^{\infty} e^{-ax^2} \sin(bx) dx &= 0 \quad (\text{odd integrand})\end{aligned}$$

**Parity / vanishing:**  $\int_{-a}^a f_{\text{odd}}(x) dx = 0, \quad \int_0^{2\pi} \sin^{2n+1} \theta d\theta = \int_0^{2\pi} \cos^{2n+1} \theta d\theta = 0.$

## 11. Fermi's Golden Rule & DOS

$$\begin{aligned}\Gamma_{i \rightarrow f} &= \frac{2\pi}{\hbar} |\langle f|H'|i\rangle|^2 \rho(E_f) \\ \rho_{3D}(E) &= \frac{Vm k}{2\pi^2 \hbar^2}, & \rho_{2D}(E) &= \frac{Am}{2\pi \hbar^2}, & \rho_{1D}(E) &= \frac{L}{\pi \hbar} \sqrt{\frac{m}{2E}}\end{aligned}$$